PHYS 357: Honours Quantum Physics 1

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Abstract

Contents

1 Introduction **1**

2 Mathematical Background **1**
2.1 Taylor series . . . 1
2.2 Euler's formula and trigonometric identities . . . 2
2.3 Bases . . . 2

3 Stern-Gerlach Experiments **2**
3.1 The classical picture . . . 2
3.1.1 Angular momentum . . . 2
3.1.2 Magnetic moment of a circulating charge . . . 3
3.1.3 Gyromagnetic ratio . . . 3
3.2 Spin . . . 3
3.2.1 Intrinsic moment and the g-factor . . . 3
3.2.2 Magnetons . . . 4
3.3 The Stern-Gerlach apparatus . . . 5
3.3.1 Force in a nonuniform field . . . 5
3.3.2 Silver atoms . . . 5
3.3.3 Classical prediction versus observation . . . 6
3.4 Spin quantization . . . 6
3.5 Sequential Stern-Gerlach experiments . . . 6
3.5.1 Experiment 1: SGz then SGz . . . 7
3.5.2 Experiment 2: SGz then SGx . . . 7
3.5.3 Experiment 3: SGz, SGx, then SGz . . . 7
3.5.4 Experiment 4: SGz, modified SGx, then SGz . . . 8

4 Quantum States **8**
4.1 Kets, bras and bases . . . 8
4.2 Measurement and probability . . . 9
4.2.1 Bras of superpositions . . . 10
4.2.2 Normalization . . . 10
4.2.3 Global and relative phase . . . 10
4.3 Expectation value and uncertainty . . . 11
4.3.1 Measured averages and shot noise . . . 11
4.4 Spin states along x and y . . . 11
4.4.1 The state $|+\hat{x}\rangle$ . . . 11
4.4.2 Experiment 5: measuring S_y . . . 12
4.4.3 Fixing the phases . . . 12

5 Matrix Mechanics **13**
5.1 States as column vectors . . . 13
5.2 Operators as matrices . . . 13
5.3 Observables as operators . . . 14
5.3.1 Spin operators and Pauli matrices . . . 15

5.3.2 Hermitian operators 16
5.4 Change of basis 16
5.4.1 Operators in another basis 18

6 Rotations **18**
6.1 The rotation operator 18
6.2 Generators of rotations 19
6.3 Functions of an operator 19
6.3.1 Rotations about $\hat{z}$ 20
6.4 Rotations about any axis 20

7 Angular Momentum **21**
7.1 Non-commuting rotations and commutation relations 21
7.2 Compatible observables 22
7.3 Uncertainty relations 23
7.4 Total angular momentum 24
7.5 Ladder operators 25
7.6 Matrix elements 26
7.7 Rotations by 2π: fermions and bosons 26
7.8 Atomic quantum numbers 27

8 Time Evolution **27**
8.1 The time-evolution operator 27
8.2 Stationary states 28
8.3 Time dependence of expectation values 29
8.4 Spin precession 29
8.4.1 Expectation values 30
8.4.2 From the equation of motion 31
8.5 Magnetic resonance 31
8.5.1 The rotating frame 32
8.5.2 Rotating-wave approximation 33
8.5.3 Rabi oscillations 33

A Solutions **34**

B Useful Links **36**

1 Introduction

Textbook used is JS Townsend, *A Modern Approach to Quantum Mechanics*, 2nd ed., University Science Books 2012. Course grading is divided as follows:

- 10% problem sets (must be handwritten scans).
- 40% quizzes (on Wednesdays, best 8 of 12).
- 20% midterms (best of 2, October 2nd and November 6th).
- 30% final exam.

2 Mathematical Background

2.1 Taylor series

Definition 1 (Taylor series)**.** *A function f that is smooth near a (it can be differentiated any number of times there) equals the power series*

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n, \tag{1}$$

where $f^{(n)}$ is the nth derivative. The special case $a = 0$ is called the Maclaurin series.

Result 1 (Series used in this course)**.**

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}, \qquad \cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}, \qquad \sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}, \tag{2}$$

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n \quad (|x|<1), \qquad (1+u)^N = \sum_{n=0}^{N} \binom{N}{n} u^n \quad (N \in \mathbb{N}), \tag{3}$$

$$\ln(1+u) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1} u^n}{n} = u - \frac{u^2}{2} + \frac{u^3}{3} - \cdots \quad (|u|<1). \tag{4}$$

Problem 1 (The exponential as a limit)**.** *Show that* $\lim_{N\to\infty} (1 + x/N)^N = e^x$.

Remark 1. *This limit is what lets a rotation be written as an exponential (Section 6.2).*

2.2 Euler's formula and trigonometric identities

Result 2 (Euler's formula)**.** *Comparing the series for $e^{i\theta}$, $\cos\theta$ and $\sin\theta$ gives $e^{i\theta} = \cos\theta + i\sin\theta$, hence*

$$\cos\theta = \frac{e^{i\theta} + e^{-i\theta}}{2}, \qquad \sin\theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}. \tag{5}$$

These are used constantly when rotation matrices are written out.

Result 3 (Double-angle identities)**.** *Squaring $e^{ix} = \cos x + i\sin x$ and comparing with $e^{2ix} = \cos 2x + i\sin 2x$,*

$$\cos^2 x - \sin^2 x = \cos 2x, \qquad 2\sin x\cos x = \sin 2x. \tag{6}$$

These turn the half angles $\phi/2$ in spin rotation matrices back into full angles ϕ.

2.3 Bases

Definition 2 (Basis)**.** *A set of vectors is* linearly independent *if none of them can be written as a combination of the others, and it* spans *a space if every vector in the space is some combination of them. A* basis *is a set that does both, so every vector is a unique combination of the basis vectors. The number of basis vectors is the dimension of the space.*

Definition 3 (Kronecker delta)**.**

$$\delta_{ij} = \begin{cases} 1 & i = j, \\ 0 & i \neq j. \end{cases} \tag{7}$$

3 Stern-Gerlach Experiments

August 31 - September 9, 2026

3.1 The classical picture

3.1.1 Angular momentum

Definition 4 (Orbital angular momentum)**.** *The orbital angular momentum of a particle is*

$$\vec{L} = \vec{r} \times \vec{p}, \tag{8}$$

where $\vec{r}$ is the position vector and $\vec{p} = m\vec{v}$ is the linear momentum.

For motion confined to the x-y plane, only the z component survives. On a circular path of radius r with speed v,

$$L_z = mvr.$$

Definition 5 (Angular momentum of a rigid body)**.** *For a rigid body*[1] *rotating about a fixed axis,*

$$L = I_0\omega,$$

with ω the angular velocity and I_0 the moment of inertia, which measures how the mass is distributed relative to the rotation axis.

[1] A rigid body is one that does not deform under the action of external forces.

3.1.2 Magnetic moment of a circulating charge

Definition 6 (Magnetic moment)**.** *A vector $\vec{\mu}$ measuring the net circulation of electric charge. It sets the torque, and in a nonuniform field the force, that a magnet experiences in an external magnetic field.*

A point particle of mass m and charge q on a circular orbit of radius r and period T passes any given point once per period, so it acts as a steady current $I = q/T = qv/2\pi r$. Its moment is

$$\mu = \frac{IA}{c}^{2} = \frac{1}{c}\left(\frac{qv}{2\pi r}\right)\left(\pi r^2\right) = \frac{q}{2mc}\, mvr = \frac{q}{2mc}\, L_z,$$

where A is the area enclosed by the orbit.

3.1.3 Gyromagnetic ratio

Definition 7 (Gyromagnetic ratio)**.** *The constant γ relating magnetic moment to angular momentum,*

$$\vec{\mu} = \gamma \vec{L}, \qquad \gamma = \frac{q}{2mc} \text{ for a circulating charge.} \tag{9}$$

Remark 2. *This is purely classical: a point charge on a definite orbit, nothing quantized. It matters because measuring $\vec{\mu}$ is how we measure angular momentum, which is what the Stern-Gerlach apparatus does.*

3.2 Spin

Definition 8 (Spin)**.** *Spin $\vec{S}$ is an* intrinsic *angular momentum: a fixed property of the particle type, like charge or mass, not caused by any motion through space. A stationary electron alone in empty space still has spin. It comes in addition to any orbital $\vec{L}$, not in place of it.*

3.2.1 Intrinsic moment and the g-factor

Spin also carries a magnetic moment proportional to it, but with an extra dimensionless factor,

$$\vec{\mu} = \gamma \vec{S}, \qquad \gamma = \frac{gq}{2mc}. \tag{10}$$

Definition 9 (g-factor)**.** *The dimensionless number g above. It says how much stronger or weaker the moment is than the classical value ($g = 1$). It must be measured or derived from relativistic quantum theory.*

[2] Gaussian (CGS) units, where a current loop has moment IA/c. In SI units it is IA.

3.2.2 Magnetons

Definition 10 (Planck constant)**.** *The constant that sets the scale at which quantization matters. It relates photon energy to frequency, $E = h\nu$, and*

$$h \approx 6.626 \times 10^{-34}\ \mathrm{J\,s}, \qquad \hbar = \frac{h}{2\pi} = 1.055 \times 10^{-27}\ \mathrm{erg\,s} = 6.582 \times 10^{-16}\ \mathrm{eV\,s}.$$

Both have units of angular momentum, $[\hbar] = [L_z] = \mathrm{kg\,m^2/s}$.

Writing $\vec{S} = \hbar(\vec{S}/\hbar)$ with $\vec{S}/\hbar$ dimensionless, the $\hbar$ joins the gyromagnetic ratio. The combination $|e|\hbar/2mc$ then carries all the units and sets the natural size of the moment for a given mass.

Definition 11 (Magnetons)**.** *The natural units of magnetic moment for the electron (Bohr magneton) and for nucleons, i.e. protons and neutrons (nuclear magneton):*

$$\mu_B = \frac{|e|\hbar}{2m_e c}, \qquad \mu_N = \frac{|e|\hbar}{2m_p c} \approx \frac{\mu_B}{1836}, \tag{11}$$

since $m_p/m_e \approx 1836$.

The magnetic moments are then

$$\vec{\mu}_e = -g_e\,\mu_B\,\frac{\vec{S}}{\hbar}, \qquad \vec{\mu}_{p,n} = g_{p,n}\,\mu_N\,\frac{\vec{S}}{\hbar}, \tag{12}$$

where the minus sign for the electron comes from its negative charge.

Particle	Charge q	Mass m	g-factor	Gyromagnetic ratio γ
Electron	$-\lvert e\rvert$	m_e	$g_e \approx 2$	$-g_e\mu_B/\hbar$
Proton	$+\lvert e\rvert$	m_p	$g_p \approx 5.59$	$g_p\mu_N/\hbar$
Neutron	0	$m_n \approx m_p$	$g_n \approx -3.83$	$g_n\mu_N/\hbar$

Table 1: Charges, masses, g-factors and gyromagnetic ratios.

Remark 3. *Classically $\mu = (q/2mc)L_z$ vanishes when $q = 0$, yet the neutral neutron has a moment. The reason is that the neutron is not a point particle: it is made of three charged quarks (one up quark with charge $+\frac{2}{3}|e|$ and two down quarks with $-\frac{1}{3}|e|$ each). The charges add to zero, but the quarks' spins and motion inside the neutron each give a moment, and these moments do not cancel. So its $\vec{\mu}$ cannot come from a single circulating point charge: a first sign the classical picture fails. The same internal structure is why $g_p \approx 5.59$ is far from 2.*

Remark 4. *Since $m_p \gg m_e$, $\mu_N \ll \mu_B$, so nuclei are deflected far less than electrons in a Stern-Gerlach device.*

3.3 The Stern-Gerlach apparatus

Definition 12 (Stern-Gerlach (SG) apparatus)**.** *A device that measures one component of a particle's magnetic moment. A beam of neutral atoms passes through a magnet shaped so that the field strength changes along a chosen axis, here z (a field* gradient*). The gradient pushes each atom by an amount proportional to μ_z, and a screen records where it lands. Since $\vec{\mu} = \gamma \vec{S}$, the landing position measures S_z.*

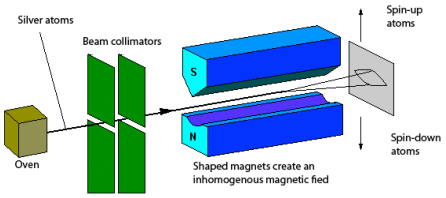


Figure 1: Illustration of the Stern-Gerlach apparatus.

3.3.1 Force in a nonuniform field

A magnetic moment in a field has potential energy $U = -\vec{\mu} \cdot \vec{B}$, so

$$\vec{F} = -\vec{\nabla} U = \vec{\nabla}(\vec{\mu} \cdot \vec{B}), \qquad F_z = \mu_z \frac{\partial B_z}{\partial z}.$$

Remark 5. *A uniform field gives no net force, only a torque (a twist that turns $\vec{\mu}$ without moving the atom). That is why the magnet needs a gradient.*

Definition 13 (Deflection)**.** *The sideways displacement of an atom on the screen caused by F_z. It is proportional to μ_z, so the screen pattern is a record of the moments in the beam.*

3.3.2 Silver atoms

Silver has zero net orbital angular momentum, $\vec{L} = 0$, and its whole moment comes from one unpaired outermost (valence) electron, so the atom behaves like a single electron spin. With

$$\mu_z = -\frac{g_e |e|}{2 m_e c} S_z$$

and a gradient $\partial B_z / \partial z < 0$,

$$F_z = g_e \frac{|e|}{2 m_e c} \left| \frac{\partial B_z}{\partial z} \right| S_z.$$

Result 4. *$F_z \propto S_z$: the deflection measures the z component of spin directly.*

3.3.3 Classical prediction versus observation

Atoms leave the oven with spins pointing in random directions, so $S_z = |\vec{S}|\cos\theta$ takes every value in $[-|\vec{S}|, |\vec{S}|]$. Classically the screen should show a continuous smear. It shows two spots.

Result 5 (Quantization of spin)**.** *The electron's spin component is quantized:*

$$S_z = \pm\frac{\hbar}{2}. \tag{13}$$

3.4 Spin quantization

Definition 14 (Spin quantum number)**.** *The number s, fixed by the particle type, taking values $s = 0, \frac{1}{2}, 1, \frac{3}{2}, \ldots$*[3] *A measurement of S_z gives*

$$S_z = m\hbar, \qquad m = -s, -s+1, \ldots, +s, \tag{14}$$

so the number of possible outcomes, and therefore of beams on the screen, is

$$n = 2s + 1. \tag{15}$$

Counting the beams gives $s = (n-1)/2$. Silver gives $n = 2$ beams, $S_z = \pm\hbar/2$, so the electron has $s = \frac{1}{2}$. The deuteron (a proton and a neutron with aligned spins) gives $n = 3$ beams, $S_z = +\hbar, 0, -\hbar$, so $s = 1$.

Remark 6. *S_z is a component. The magnitude is $|\vec{S}| = \hbar\sqrt{s(s+1)}$, which for the electron ($s = 1/2$) is $\hbar\sqrt{3}/2$.*

3.5 Sequential Stern-Gerlach experiments

We write SGz for a device whose gradient points along z, and SGx for the same device rotated by 90° about the beam axis.

Definition 15 (Ket)**.** *A ket $|\cdot\rangle$ is a label for a quantum state, treated as a vector. For electron spin, the two outcomes of an SGz give*

$$|+\hat{z}\rangle \equiv |\uparrow\rangle \text{ for } S_z = +\frac{\hbar}{2}, \qquad |-\hat{z}\rangle \equiv |\downarrow\rangle \text{ for } S_z = -\frac{\hbar}{2},$$

and likewise $|\pm\hat{x}\rangle$ for $S_x = \pm\hbar/2$, and so on for any axis. We follow Townsend and use the $|\pm\hat{n}\rangle$ labels.

Definition 16 (Filter)**.** *Each device sends atoms into two beams. A filter blocks one of them.*

Remark 7. *A beam straight from the oven always splits 50/50 in any SG device, whatever its orientation.*

[3] They follow from $[J_x, J_y] = i\hbar J_z$ via the ladder operators (Section 7.5), which force m to step by 1 from $-s$ to $+s$, so $2s$ is a whole number.

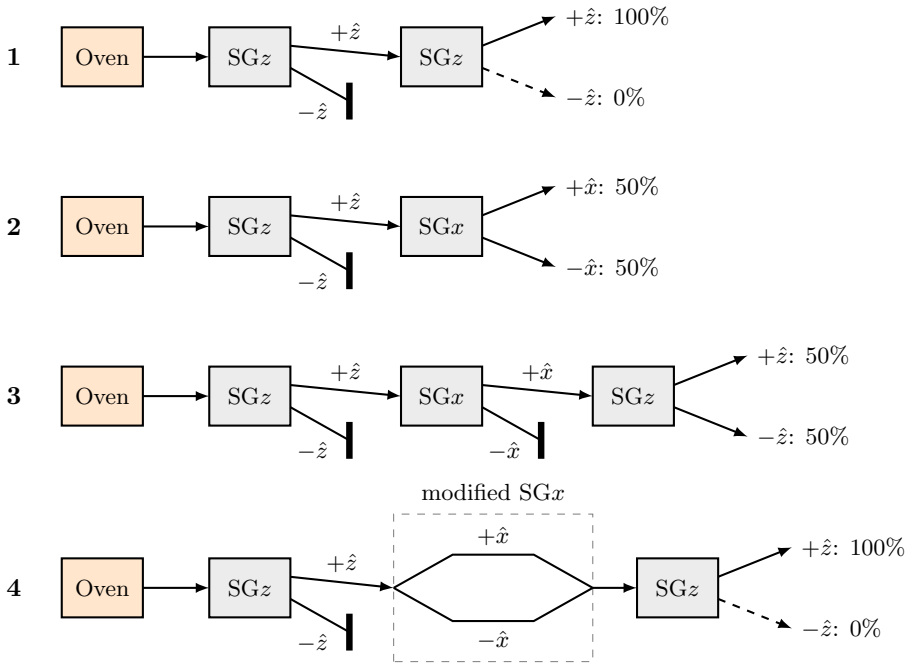


Figure 2: Sequential Stern-Gerlach experiments. Thick bars block a beam. Percentages are the fractions of atoms entering the last device. Every oven starts with N_0 atoms.

Remark 8. *The lecture notes use N_0 for the number of atoms entering the* second *device rather than leaving the oven, so their counts are twice the ones below.*

3.5.1 Experiment 1: SGz then SGz

Filter the $+\hat{z}$ beam and send it into a second SGz. Every atom comes out $+\hat{z}$. The measurement is repeatable: an atom found with $S_z = +\hbar/2$ gives the same answer again, so it left the first device in $|+\hat{z}\rangle$. We end with $N_0/2$ atoms in $|+\hat{z}\rangle$.

3.5.2 Experiment 2: SGz then SGx

Filter $+\hat{z}$ and send it into an SGx. Half come out $+\hat{x}$, half $-\hat{x}$: knowing S_z tells us nothing about S_x. We end with $N_0/4$ in each of $|\pm\hat{x}\rangle$.

3.5.3 Experiment 3: SGz, SGx, then SGz

Filter $+\hat{z}$, filter $+\hat{x}$, then measure S_z again. The beam splits 50/50 even though every atom was $+\hat{z}$ at the start. Blocking $+\hat{x}$ instead gives the same

result. We end with $N_0/8$ in each of $|\pm\hat{z}\rangle$.

Result 6. *No state has definite values of both S_x and S_z. Measuring S_x destroys what we knew about S_z.*

3.5.4 Experiment 4: SGz, modified SGx, then SGz

Definition 17 (Modified SG device)**.** *An SG device that splits the beam according to S_n and then recombines the two beams, so both outcomes exit at the same place without being recorded.*

Filter $+\hat{z}$, pass it through a modified SGx with neither path blocked, then measure S_z. Every atom comes out $+\hat{z}$: all $N_0/2$ end in $|+\hat{z}\rangle$.

Result 7. *The atoms cannot be going half through $+\hat{x}$ and half through $-\hat{x}$, since Experiment 3 showed each half alone gives 50/50. Instead the two paths* interfere*: as with water or light waves, contributions from the two paths add and can cancel, here cancelling the $-\hat{z}$ outcome. No definite path can be assigned to an atom. Section 4 makes this precise.*

Remark 9. *With one path blocked, a modified SG acts exactly like an ordinary SG.*

4 Quantum States

September 11-18, 2026

Throughout, an *n-level system* is one whose measurements have n possible outcomes. For spin s, $n = 2s + 1$, so spin-$\frac{1}{2}$ has $n = 2$.

4.1 Kets, bras and bases

Definition 18 (Bra and inner product)**.** *Every ket (Def. 15) $|\psi\rangle$ has a matching bra $\langle\psi|$, its conjugate transpose (adjoint): the ket's coefficients are complex conjugated, and the column becomes a row (Section 4.2.1). A bra followed by a ket gives a complex number, the inner product $\langle\phi|\psi\rangle \in \mathbb{C}$.*

Definition 19 (Orthonormal basis)**.** *The outcome states $\{|j\rangle : j = 1, \ldots, n\}$ of one measurement form a basis that is orthonormal:*

$$\langle i|j\rangle = \delta_{ij}. \tag{16}$$

For spin-$\frac{1}{2}$ this is $\langle\pm\hat{z}|\pm\hat{z}\rangle = 1$ and $\langle\pm\hat{z}|\mp\hat{z}\rangle = 0$, and the same holds for $\{|\pm\hat{x}\rangle\}$ or any other axis.

Remark 10. *The label j counts* outcomes*, not axes. Choosing to measure S_x gives $|1\rangle = |+\hat{x}\rangle$, $|2\rangle = |-\hat{x}\rangle$; choosing S_z gives $|1\rangle = |+\hat{z}\rangle$, $|2\rangle = |-\hat{z}\rangle$. Either way $n = 2$. A ket may also carry several labels, usually the quantum numbers that pin it down: $|s, m\rangle$ for spin, so $|\pm\hat{z}\rangle = |\frac{1}{2}, \pm\frac{1}{2}\rangle$, and for the deuteron $|1, +1\rangle, |1, 0\rangle, |1, -1\rangle$. Electrons in atoms are labelled the same way (Section 7.8).*

Definition 20 (Superposition)**.** *Any state is a linear combination of basis kets,*

$$|\psi\rangle = \sum_{j=1}^{n} c_j \, |j\rangle \,, \qquad c_j \in \mathbb{C}; \qquad \textit{spin-}\tfrac{1}{2}: \; |\psi\rangle = c_+ \, |+\hat{z}\rangle + c_- \, |-\hat{z}\rangle \,. \tag{17}$$

Proposition 1 (Coefficients are inner products)**.**

$$c_j = \langle j|\psi\rangle \,. \tag{18}$$

Proof. $\langle i|\psi\rangle = \sum_j c_j \langle i|j\rangle = \sum_j c_j \delta_{ij} = c_i$. □

Definition 21 (Projector)**.** *The outer product (a ket times a bra)* $P_j = |j\rangle \langle j|$*. It is an operator that keeps only the* $|j\rangle$ *part of a state,*

$$P_j \, |\psi\rangle = |j\rangle \langle j|\psi\rangle = c_j \, |j\rangle \,,$$

and projecting twice changes nothing: $P_j^2 = |j\rangle \langle j|j\rangle \langle j| = P_j$*.*

Proposition 2 (Completeness relation)**.** *Because every state can be expanded in the basis, the projectors sum to the identity:*

$$\mathbb{1} = \sum_{j=1}^{n} |j\rangle \langle j| \,; \qquad \textit{spin-}\tfrac{1}{2}: \; \mathbb{1} = |+\hat{n}\rangle \langle +\hat{n}| + |-\hat{n}\rangle \langle -\hat{n}| \;\; \textit{for any axis } \hat{n}. \tag{19}$$

Remark 11 (Inserting the identity)**.** *Writing* $\mathbb{1}$ *in this form in front of a state reproduces its expansion,* $|\psi\rangle = \sum_j |j\rangle \langle j|\psi\rangle$*. This trick is used in almost every derivation below.*

4.2 Measurement and probability

Definition 22 (Observable)**.** *A measurable quantity A. Its possible outcomes* a_j *each come with a state* $|a_j\rangle$*, and these states form an orthonormal basis. For spin-*$\frac{1}{2}$*, A can be* S_x*,* S_y *or* S_z*, with outcomes* $\pm\hbar/2$*.*

Principle 1 (Born rule)**.** *The probability that a system in* $|\psi\rangle$ *is found in* $|\phi\rangle$ *is*

$$P(\phi \,|\, \psi) = |\langle \phi|\psi\rangle|^2 \in [0, 1]. \tag{20}$$

In particular $P(A = a_j) = |\, \langle a_j|\psi\rangle \,|^2$*, e.g.* $P(S_z = \pm\hbar/2) = |c_\pm|^2$*.*

Definition 23 (Probability amplitude)**.** *The complex number* $\langle \phi|\psi\rangle$ *whose modulus squared is the probability.*

Remark 12. *The Born rule has no deeper derivation: it is a postulate that agrees with every experiment. As David Mermin put it, "shut up and calculate."*

4.2.1 Bras of superpositions

Write the bra with unknown coefficients, $\langle\psi| = c'_+ \langle+\hat{z}| + c'_- \langle-\hat{z}|$. By orthonormality,

$$\langle\psi|\psi\rangle = c'_+ c_+ + c'_- c_-.$$

A system in $|\psi\rangle$ is found in $|\psi\rangle$ with certainty, and it is always found in one of the two beams, so this must equal $|c_+|^2 + |c_-|^2 = 1$. That forces $c'_\pm = c^*_\pm$.

Result 8. *Going from a ket to its bra, conjugate the coefficients:*

$$\langle\psi| = \sum_j c_j^* \langle j|, \qquad \langle\phi|\psi\rangle = \langle\psi|\phi\rangle^*. \tag{21}$$

4.2.2 Normalization

Proposition 3 (Normalization)**.** $\langle\psi|\psi\rangle = 1$ *is the statement that the probabilities of all outcomes add to 1.*

Proof.

$$\langle\psi|\psi\rangle = \sum_{i,j} c_i^* c_j \langle a_i|a_j\rangle = \sum_j |c_j|^2 = \sum_j P(A = a_j) = 1.$$

□

Example. *For* $|\psi\rangle = \frac{1}{\sqrt{3}}|+\hat{z}\rangle + i\sqrt{\frac{2}{3}}|-\hat{z}\rangle$, $c_+ = 1/\sqrt{3}$ *and* $c_- = i\sqrt{2/3}$, *so* $P(S_z = +\hbar/2) = 1/3$ *and* $P(S_z = -\hbar/2) = |i|^2 \cdot 2/3 = 2/3$. *They add to 1, so the state is normalized.*

4.2.3 Global and relative phase

Definition 24 (Global and relative phase)**.** *A* global *phase multiplies the whole ket:* $e^{i\alpha}|\psi\rangle$ *with* $\alpha \in \mathbb{R}$. *A* relative *phase sits between the terms of a superposition, like the* $e^{i\delta}$ *in* $c_+|+\hat{z}\rangle + e^{i\delta}c_-|-\hat{z}\rangle$, *and cannot be pulled out front.*

Proposition 4. *A global phase changes no probability:* $\left|\langle\phi|e^{i\alpha}\psi\rangle\right|^2 = |e^{i\alpha}|^2 |\langle\phi|\psi\rangle|^2 = |\langle\phi|\psi\rangle|^2$ *for every* $|\phi\rangle$.

Definition 25 (Physical state)**.** *What the system is actually like, as far as any experiment can tell: the probabilities of every outcome of every possible measurement. Two kets that give the same probabilities for all measurements describe the same physical state.*

Result 9. $|\psi\rangle$ *and* $e^{i\alpha}|\psi\rangle$ *are the same physical state, so global phases can be dropped. Relative phases cannot: they leave the* S_z *probabilities alone but change the* S_x *and* S_y *ones (Section 4.4).*

4.3 Expectation value and uncertainty

Definition 26 (Expectation value)**.** *The average outcome, weighted by the probabilities:*

$$\langle A\rangle = \sum_j a_j\, P(A=a_j) = \sum_j a_j\, |\langle a_j|\psi\rangle|^2\,; \qquad \langle S_z\rangle = \frac{\hbar}{2}|c_+|^2 - \frac{\hbar}{2}|c_-|^2. \quad (22)$$

Definition 27 (Uncertainty)**.** *The standard deviation (typical spread of outcomes around the average):*

$$(\Delta A)^2 = \langle (A-\langle A\rangle)^2\rangle = \langle A^2\rangle - \langle A\rangle^2\,, \qquad \langle A^2\rangle = \sum_j a_j^2\, |\langle a_j|\psi\rangle|^2\,. \quad (23)$$

Remark 13. *For spin-$\frac{1}{2}$ both outcomes square to $\hbar^2/4$, so $\langle S_z^2\rangle = \hbar^2/4$ for every state.*

Example. *For $|+\hat{x}\rangle$, Experiment 3 gives both S_z probabilities $1/2$, so $\langle S_z\rangle = 0$ and $\Delta S_z = \hbar/2$. The average is zero but every single measurement gives $\pm\hbar/2$: maximal spread.*

Example. *For $|\psi\rangle = \sqrt{\frac{1}{3}}\,|+\hat{z}\rangle + i\sqrt{\frac{2}{3}}\,|-\hat{z}\rangle$,*

$$\langle S_z\rangle = \frac{\hbar}{2}\cdot\frac{1}{3} - \frac{\hbar}{2}\cdot\frac{2}{3} = -\frac{\hbar}{6}, \qquad (\Delta S_z)^2 = \frac{\hbar^2}{4} - \frac{\hbar^2}{36} = \frac{2}{9}\hbar^2, \qquad \Delta S_z = \frac{\sqrt{2}}{3}\hbar.$$

4.3.1 Measured averages and shot noise

Remark 14. *For N measurements on identically prepared atoms, with $N_\pm$ giving $\pm\hbar/2$, the measured average is $\langle S_z\rangle_N = \frac{\hbar}{2}\frac{N_+ - N_-}{N}$, which tends to $\langle S_z\rangle$ as $N\to\infty$.*

Remark 15 (Shot noise)**.** *The measured average $\langle S_z\rangle_N$ scatters around the true expectation value, and its average deviation from it is*

$$|\langle S_z\rangle_N - \langle S_z\rangle| \sim \frac{\Delta S_z}{\sqrt{N}}.$$

This scatter is called shot noise. It comes from the randomness of individual quantum outcomes, not from a faulty apparatus, and is absent only when $\Delta S_z = 0$, i.e. when the state has a definite S_z.

4.4 Spin states along x and y

4.4.1 The state $|+\hat{x}\rangle$

Experiment 3 gives $|\,\langle\pm\hat{z}| + \hat{x}\rangle\,|^2 = \frac{1}{2}$, so both amplitudes have modulus $1/\sqrt{2}$ with their own phases:

$$|+\hat{x}\rangle = \frac{e^{i\delta_+}}{\sqrt{2}}\left(|+\hat{z}\rangle + e^{i\delta}\,|-\hat{z}\rangle\right), \qquad \delta = \delta_- - \delta_+.$$

Dropping the global phase $e^{i\delta_+}$,

$$|+\hat{x}\rangle = \frac{1}{\sqrt{2}}\left(|+\hat{z}\rangle + e^{i\delta}\,|-\hat{z}\rangle\right).$$

Experiment 3 alone does not fix the relative phase δ.

4.4.2 Experiment 5: measuring S_y

Definition 28 (Cyclic permutation)**.** *Relabelling the axes $x \to y \to z \to x$ (or the reverse). It is a rotation of the coordinate system, so no physics changes.*

Experiment 5 is Experiment 3 cyclically permuted: filter $+\hat{z}$, then measure S_y. By symmetry it also splits 50/50, so every cross-axis overlap has modulus squared $\frac{1}{2}$:

$$|\langle +\hat{x}|+\hat{z}\rangle|^2 = |\langle +\hat{y}|+\hat{z}\rangle|^2 = |\langle -\hat{x}|+\hat{y}\rangle|^2 = \frac{1}{2}.$$

The same argument as for $|+\hat{x}\rangle$ gives, after dropping global phases,

$$|\pm\hat{x}\rangle = \frac{1}{\sqrt{2}}\left(|+\hat{z}\rangle \pm e^{i\delta}\,|-\hat{z}\rangle\right), \qquad |\pm\hat{y}\rangle = \frac{1}{\sqrt{2}}\left(|+\hat{z}\rangle \pm e^{i\gamma}\,|-\hat{z}\rangle\right).$$

4.4.3 Fixing the phases

Using orthonormality of the $\hat{z}$ basis,

$$|\langle +\hat{x}|+\hat{y}\rangle|^2 = \left|\frac{1}{2}\left(1+e^{i(\gamma-\delta)}\right)\right|^2 = \frac{1}{2}\left(1+\cos(\gamma-\delta)\right).$$

Setting this to $\frac{1}{2}$ forces $\gamma - \delta = \pm\pi/2$. The rest is convention.

Result 10 (Phase convention)**.** *With $\delta = 0$ and $\gamma = +\pi/2$,*

$$|\pm\hat{x}\rangle = \frac{1}{\sqrt{2}}\left(|+\hat{z}\rangle \pm |-\hat{z}\rangle\right), \qquad |\pm\hat{y}\rangle = \frac{1}{\sqrt{2}}\left(|+\hat{z}\rangle \pm i\,|-\hat{z}\rangle\right), \tag{24}$$

$$\langle\pm\hat{x}| = \frac{1}{\sqrt{2}}\left(\langle+\hat{z}| \pm \langle-\hat{z}|\right), \qquad \langle\pm\hat{y}| = \frac{1}{\sqrt{2}}\left(\langle+\hat{z}| \mp i\,\langle-\hat{z}|\right). \tag{25}$$

Remark 16. *Complex coefficients are required: with real ones the relative phases could only be 0 or π, and $\gamma - \delta = \pm\pi/2$ would be impossible.*

Example. *For $|\psi\rangle = \sqrt{\frac{1}{3}}\,|+\hat{z}\rangle + i\sqrt{\frac{2}{3}}\,|-\hat{z}\rangle$, compute $\langle S_y\rangle$. Using $\langle\pm\hat{y}|+\hat{z}\rangle = \frac{1}{\sqrt{2}}$ and $\langle\pm\hat{y}|-\hat{z}\rangle = \mp\frac{i}{\sqrt{2}}$,*

$$\langle\pm\hat{y}|\psi\rangle = \frac{1}{\sqrt{2}}\left(\sqrt{\tfrac{1}{3}} \pm \sqrt{\tfrac{2}{3}}\right), \qquad P\big(S_y = \pm\tfrac{\hbar}{2}\big) = \frac{1}{2} \pm \frac{\sqrt{2}}{3},$$

which add to 1. Then

$$\langle S_y\rangle = \frac{\hbar}{2}\left[\left(\frac{1}{2}+\frac{\sqrt{2}}{3}\right) - \left(\frac{1}{2}-\frac{\sqrt{2}}{3}\right)\right] = \frac{\sqrt{2}}{3}\hbar.$$

The factor i did nothing for S_z, but as a relative phase it is exactly what tilts the spin toward $+\hat{y}$.

5 Matrix Mechanics

September 21-25, 2026

5.1 States as column vectors

Definition 29 (Representation of a state)**.** *In a basis, a ket is represented by the column of its coefficients and its bra by the conjugate row (the adjoint, †). In the S_z basis:*

$$|\psi\rangle \xrightarrow{S_z} \psi = \begin{pmatrix} c_+ \\ c_- \end{pmatrix}, \qquad \langle\psi| \xrightarrow{S_z} \psi^\dagger = \begin{pmatrix} c_+^* & c_-^* \end{pmatrix}, \qquad c_\pm = \langle \pm\hat{z}|\psi\rangle . \tag{26}$$

Remark 17. *Write $|\psi\rangle \xrightarrow{S_z} \psi$, not $|\psi\rangle = \psi$: the ket is basis-independent, the column is not.*

Example.

$$|+\hat{z}\rangle \xrightarrow{S_z} \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad |-\hat{z}\rangle \xrightarrow{S_z} \begin{pmatrix} 0 \\ 1 \end{pmatrix},$$

$$|\pm\hat{x}\rangle \xrightarrow{S_z} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ \pm 1 \end{pmatrix}, \qquad |\pm\hat{y}\rangle \xrightarrow{S_z} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ \pm i \end{pmatrix}.$$

Problem 2 (Other bases)**.** *In the S_x basis, $|\psi\rangle \xrightarrow{S_x} \begin{pmatrix} \langle +\hat{x}|\psi\rangle \\ \langle -\hat{x}|\psi\rangle \end{pmatrix}$, and similarly for S_y. Write the column representing $|+\hat{z}\rangle$ in the S_z, S_x and S_y bases.*

Proposition 5 (Inner product as a matrix product)**.** *If $|\phi\rangle \xrightarrow{S_z} \phi$ and $|\psi\rangle \xrightarrow{S_z} \psi$, then $\langle\phi|\psi\rangle = \phi^\dagger\psi$ (row times column), so $P(\phi\,|\,\psi) = \left|\phi^\dagger\psi\right|^2$ and $\langle\psi|\psi\rangle = \psi^\dagger\psi$.*

Remark 18. *For a state given in another basis, replace each basis ket by its S_z column and add. For $|\psi\rangle = a\,|+\hat{y}\rangle + b\,|-\hat{y}\rangle$,*

$$\psi \xrightarrow{S_z} \frac{a}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix} + \frac{b}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} a+b \\ i(a-b) \end{pmatrix}.$$

Checking $\psi^\dagger\psi = 1$ before going further catches most mistakes. A cleaner systematic method is in Section 5.4.

Problem 3 (A probability by matrix mechanics)**.** *Using matrix mechanics, find $P(S_x = +\hbar/2)$ for the state $|\psi\rangle = \sqrt{\frac{1}{5}}\,|+\hat{z}\rangle + i\sqrt{\frac{4}{5}}\,|-\hat{z}\rangle$.*

5.2 Operators as matrices

Definition 30 (Linear operator)**.** *A map A from kets to kets that respects superpositions, $A\,(a\,|\psi\rangle + b\,|\phi\rangle) = aA\,|\psi\rangle + bA\,|\phi\rangle$. Every operator in this course is linear.*

Definition 31 (Matrix of an operator)**.** *In a basis $\{|j\rangle\}$, A is represented by the matrix of* matrix elements

$$A_{ij} = \langle i|\,A\,|j\rangle . \tag{27}$$

Proposition 6 (Operator acting on a state)**.** *$A|\psi\rangle$ is represented by the matrix times the column, $A\psi$.*

Proof. The kth entry of the column for $A|\psi\rangle$ is $\langle k|A|\psi\rangle$. Inserting the identity,

$$\langle k|A|\psi\rangle = \sum_j \langle k|A|j\rangle\langle j|\psi\rangle = \sum_j A_{kj}\, c_j,$$

which is row k of A times the column ψ. □

Example. *The projectors and identity in the S_z basis:*

$$P_{+\hat{z}} \xrightarrow{S_z} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \qquad P_{-\hat{z}} \xrightarrow{S_z} \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \qquad \mathbb{1} \xrightarrow{S_z} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix},$$

and indeed $P_{+\hat{z}}\psi = (c_+, 0)^T$.

Definition 32 (Adjoint)**.** *The adjoint $A^\dagger$ is the conjugate transpose: $(A^\dagger)_{ij} = A^*_{ji}$. In operator form, writing $A = \sum_{i,j} A_{ij}|i\rangle\langle j|$,*

$$A^\dagger = \sum_{i,j} A^*_{ij}|j\rangle\langle i|. \tag{28}$$

For real entries it is just the transpose.

Proposition 7 (Bra of a transformed state)**.** *If $|\psi'\rangle = A|\psi\rangle$, then $\langle\psi'| = \langle\psi|A^\dagger$.*

Proof. The column of $|\psi'\rangle$ is $A\psi$, and its bra is the conjugate transpose, $(A\psi)^\dagger = \psi^\dagger A^\dagger$. □

Proposition 8 (Adjoint of a product)**.**

$$(AB)^\dagger = B^\dagger A^\dagger. \tag{29}$$

The order reverses; $A^\dagger B^\dagger$ is a different matrix in general.

5.3 Observables as operators

Definition 33 (Eigenstate and eigenvalue)**.** *$|a\rangle$ is an eigenstate of A with eigenvalue a if*

$$A|a\rangle = a|a\rangle, \tag{30}$$

the familiar $Mv = \lambda v$ in matrix form. A general operator turns a state into a different state; eigenstates come back as themselves times a number. A basis made of eigenstates of A is called an eigenbasis *of A.*

Remark 19 (Eigenstate vs eigenvector)**.** *Same thing in two languages. "Eigenstate" is the physics word for the ket, e.g. $|+\hat{x}\rangle$. "Eigenvector" is the linear-algebra word, usually for its column in some basis, e.g. $\frac{1}{\sqrt{2}}(1,1)^T$ in the S_z basis.*

Proposition 9 (Operator of an observable)**.** *For an observable with outcomes a_j and states $|a_j\rangle$, define*

$$\hat{A} = \sum_j a_j \, |a_j\rangle \, \langle a_j| \, . \tag{31}$$

Then $\hat{A}\,|a_j\rangle = a_j\,|a_j\rangle$ (the outcomes are its eigenvalues, the outcome states its eigenstates), and

$$\langle A\rangle = \langle\psi|\,\hat{A}\,|\psi\rangle\,, \qquad \langle A^2\rangle = \langle\psi|\,\hat{A}^2\,|\psi\rangle\,. \tag{32}$$

Proof. $\hat{A}\,|a_k\rangle = \sum_j a_j\,|a_j\rangle\,\delta_{jk} = a_k\,|a_k\rangle$, and $\langle\psi|\,\hat{A}\,|\psi\rangle = \sum_j a_j\,\langle\psi|a_j\rangle\,\langle a_j|\psi\rangle = \sum_j a_j\,|\langle a_j|\psi\rangle|^2 = \langle A\rangle$. Likewise $\hat{A}^2 = \sum_j a_j^2\,|a_j\rangle\,\langle a_j|$. □

Proposition 10 (An operator is diagonal in its own eigenbasis)**.**

$$\langle a_i|\,\hat{A}\,|a_j\rangle = a_j\delta_{ij}, \qquad \hat{A} \xrightarrow{|a_j\rangle} D = \begin{pmatrix} a_1 & & 0 \\ & \ddots & \\ 0 & & a_n \end{pmatrix}. \tag{33}$$

5.3.1 Spin operators and Pauli matrices

Definition 34 (Spin operators)**.** *Applying the operator of an observable to spin, for $\hat{n} \in \{\hat{x}, \hat{y}, \hat{z}\}$,*

$$\hat{S}_n = \frac{\hbar}{2}\left(|+\hat{n}\rangle\,\langle+\hat{n}| - |-\hat{n}\rangle\,\langle-\hat{n}|\right), \qquad \hat{S}_n\,|\pm\hat{n}\rangle = \pm\frac{\hbar}{2}\,|\pm\hat{n}\rangle\,. \tag{34}$$

Example. *Each spin operator in its own eigenbasis, using $\hat{S}_n\,|\pm\hat{n}\rangle = \pm\frac{\hbar}{2}\,|\pm\hat{n}\rangle$ and orthonormality:*

$$\langle\pm\hat{n}|\,\hat{S}_n\,|\pm\hat{n}\rangle = \pm\frac{\hbar}{2}, \qquad \langle\pm\hat{n}|\,\hat{S}_n\,|\mp\hat{n}\rangle = \mp\frac{\hbar}{2}\,\langle\pm\hat{n}|\mp\hat{n}\rangle = 0,$$

so

$$\hat{S}_x \xrightarrow{S_x} \frac{\hbar}{2}\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}, \qquad \hat{S}_y \xrightarrow{S_y} \frac{\hbar}{2}\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}, \qquad \hat{S}_z \xrightarrow{S_z} \frac{\hbar}{2}\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}.$$

In the S_z basis the matrix elements of $\hat{S}_n$ are

$$\langle i|\,\hat{S}_n\,|j\rangle = \frac{\hbar}{2}\Big(\,\langle i| + \hat{n}\rangle\,\langle+\hat{n}|j\rangle - \langle i| - \hat{n}\rangle\,\langle-\hat{n}|j\rangle\,\Big), \qquad i,j \in \{+\hat{z}, -\hat{z}\}.$$

Problem 4 (The matrix of $\hat{S}_z$)**.** *Find the matrix representation of $\hat{S}_z$ in the S_z basis.*

Problem 5 (The matrix of $\hat{S}_x$)**.** *Find the matrix representation of $\hat{S}_x$ in the S_z basis.*

Problem 6 (The matrix of $\hat{S}_y$)**.** *Find the matrix representation of $\hat{S}_y$ in the S_z basis.*

Remark 20 (Shortcut)**.** *Write each projector as a column times its conjugate row, e.g.*

$$|\pm\hat{x}\rangle\langle\pm\hat{x}| \xrightarrow{S_z} \frac{1}{2}\begin{pmatrix}1\\ \pm1\end{pmatrix}\begin{pmatrix}1 & \pm1\end{pmatrix} = \frac{1}{2}\begin{pmatrix}1 & \pm1\\ \pm1 & 1\end{pmatrix},$$

then subtract and multiply by $\hbar/2$.

Definition 35 (Pauli matrices)**.** *The results, all in the S_z basis:*

$$\hat{S}_j = \frac{\hbar}{2}\sigma_j, \qquad \sigma_x = \begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}, \quad \sigma_y = \begin{pmatrix}0 & -i\\ i & 0\end{pmatrix}, \quad \sigma_z = \begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}. \tag{35}$$

Each equals its own adjoint and squares to $\mathbb{1}$, so $\hat{S}_j^2 = \frac{\hbar^2}{4}\mathbb{1}$. Collected into the vector of Pauli matrices $\vec{\sigma} = \sigma_x\hat{x} + \sigma_y\hat{y} + \sigma_z\hat{z}$, this reads $\vec{S} \xrightarrow{S_z} \frac{\hbar}{2}\vec{\sigma}$.

Problem 7 (Expectation value and uncertainty by matrix mechanics)**.** *For $|\psi\rangle = \sqrt{\frac{1}{5}}\,|+\hat{y}\rangle + i\sqrt{\frac{4}{5}}\,|-\hat{y}\rangle$, find $\langle S_z\rangle$ and ΔS_z using matrix mechanics.*

5.3.2 Hermitian operators

Definition 36 (Hermitian operator)**.** *A is Hermitian if it equals its own adjoint, $A = A^\dagger$ (e.g. the Pauli matrices).*

Principle 2. *Every observable is represented by a Hermitian operator.*

Proposition 11. *The eigenvalues of a Hermitian operator are real, so measurement outcomes are real.*

Proof. If $A|a\rangle = a|a\rangle$, the bra is $\langle a|A^\dagger = a^*\langle a|$, so $a\langle a|a\rangle = \langle a|A|a\rangle = \langle a|A^\dagger|a\rangle = a^*\langle a|a\rangle$, and $a = a^*$. □

5.4 Change of basis

Take an n-level system with two orthonormal bases, $\{|j\rangle\}$ and $\{|\alpha\rangle\}$, $j, \alpha = 1, \ldots, n$. Within each basis the overlaps are Kronecker deltas, but across bases they are not: $\langle+\hat{x}|+\hat{z}\rangle = \frac{1}{\sqrt{2}}$ is neither 0 nor 1. The same ket then has two columns,

$$|\psi\rangle \xrightarrow{|j\rangle} \psi^j = \begin{pmatrix}\langle 1|\psi\rangle\\ \vdots\\ \langle n|\psi\rangle\end{pmatrix}, \qquad |\psi\rangle \xrightarrow{|\alpha\rangle} \psi^\alpha = \begin{pmatrix}\langle \alpha{=}1|\psi\rangle\\ \vdots\\ \langle \alpha{=}n|\psi\rangle\end{pmatrix}.$$

Inserting $\mathbb{1} = \sum_\alpha |\alpha\rangle\langle\alpha|$ into each $|j\rangle$ amplitude,

$$\langle j|\psi\rangle = \sum_\alpha \langle j|\alpha\rangle\langle\alpha|\psi\rangle.$$

Result 11 (Change-of-basis matrix)**.**

$$\psi^j = U\psi^\alpha, \qquad \psi^\alpha = U^\dagger\psi^j, \qquad U_{j\alpha} = \langle j|\alpha\rangle\,. \tag{36}$$

The columns of U are the new basis kets $|\alpha\rangle$ written in the old basis $\{|j\rangle\}$.

Definition 37 (Unitary operator)**.** *U is unitary if $U^\dagger U = UU^\dagger = \mathbb{1}$, i.e. its inverse is its adjoint.*

Theorem 1. *The change-of-basis matrix U is unitary.*

Proof. With $(U^\dagger)_{\alpha j} = \langle j|\alpha\rangle^* = \langle\alpha|j\rangle$ and completeness of $\{|j\rangle\}$,

$$(U^\dagger U)_{\alpha\beta} = \sum_j \langle\alpha|j\rangle\langle j|\beta\rangle = \langle\alpha|\beta\rangle = \delta_{\alpha\beta}.$$

Completeness of $\{|\alpha\rangle\}$ gives $UU^\dagger = \mathbb{1}$ the same way. □

Remark 21. *So normalization does not depend on the basis: $\psi^{j\dagger}\psi^j = \psi^{\alpha\dagger}U^\dagger U\psi^\alpha = \psi^{\alpha\dagger}\psi^\alpha$.*

Remark 22. *If a unitary U maps $|\psi\rangle$ to $c\,|\phi\rangle$ with $|\phi\rangle$ normalized, then $1 = \langle\psi|\,U^\dagger U\,|\psi\rangle = |c|^2$, so $c = e^{i\varphi}$. In particular, eigenvalues of a unitary are pure phases, and U leaves a physical state unchanged only when that state is an eigenstate of U.*

Example (Between S_x and S_z)**.** *With $|j\rangle = |\pm\hat{z}\rangle$ and $|\alpha\rangle = |\pm\hat{x}\rangle$, the columns of U are the S_z columns of $|\pm\hat{x}\rangle$:*

$$U = \begin{pmatrix} \langle+\hat{z}|+\hat{x}\rangle & \langle+\hat{z}|-\hat{x}\rangle \\ \langle-\hat{z}|+\hat{x}\rangle & \langle-\hat{z}|-\hat{x}\rangle \end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}. \tag{37}$$

The two directions are

$$S_x \to S_z:\ \psi^z = U\psi^x, \qquad S_z \to S_x:\ \psi^x = U^\dagger\psi^z = U\psi^z,$$

since U is real and symmetric, so $U^\dagger = U$: the same matrix goes both ways ($U^2 = \mathbb{1}$). Bill calls U "$R\left(\frac{\pi}{2}\hat{y}\right)$", but it is not the actual rotation matrix (see the warning in Section 6.4).

Remark 23. *Between S_y and S_z the matrix has columns $\frac{1}{\sqrt{2}}(1,\pm i)^T$,*

$$U_y = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ i & -i \end{pmatrix}, \qquad \psi^z = U_y\psi^y, \qquad \psi^y = U_y^\dagger\psi^z,$$

and here $U_y^\dagger \neq U_y$, so the reverse direction needs the actual adjoint.

Problem 8 (Expectation value from the S_x basis)**.** *For $|\psi\rangle = \frac{1}{\sqrt{3}}\,|+\hat{x}\rangle + i\sqrt{\frac{2}{3}}\,|-\hat{x}\rangle$, find $\langle S_z\rangle = \psi^{z\dagger}\hat{S}_z\psi^z$ using matrix mechanics.*

5.4.1 Operators in another basis

Let $\{|j\rangle\} = \{|a_j\rangle\}$ be the eigenbasis of $\hat{A}$, so $\hat{A} \xrightarrow{|a_j\rangle} D$ (diagonal), and $U_{j\alpha} = \langle a_j|\alpha\rangle$.

Proposition 12 (Similarity transformation)**.**

$$\hat{A} \xrightarrow{|\alpha\rangle} A^{\alpha} = U^{\dagger} D\, U. \tag{38}$$

Rewriting the same operator in a new basis this way is called a similarity transformation.

Proof. Insert the identity twice:

$$\langle\alpha|\,\hat{A}\,|\beta\rangle = \sum_{i,j} \langle\alpha|a_i\rangle\,\langle a_i|\,\hat{A}\,|a_j\rangle\,\langle a_j|\beta\rangle = \sum_{i,j} (U^{\dagger})_{\alpha i} D_{ij} U_{j\beta} = (U^{\dagger} D\, U)_{\alpha\beta}.$$

□

Result 12 (Expectation value in any basis)**.**

$$\langle A\rangle = \psi^{j\dagger} D\, \psi^{j} = \psi^{\alpha\dagger} \left[U^{\dagger} D\, U\right] \psi^{\alpha}. \tag{39}$$

Either move the state to the eigenbasis, or move the operator to the state's basis. Both give the same number.

Problem 9 ($\hat{S}_z$ in the S_x basis)**.** *Using U from Eq. (37), find the matrix of $\hat{S}_z$ in the S_x basis, and use it to redo the problem "Expectation value from the S_x basis" without changing the state's basis.*

6 Rotations

September 21-28, 2026

6.1 The rotation operator

Definition 38 (Rotation operator)**.** *$R(\theta\hat{n})$ is the linear operator that rotates a state by angle θ about the axis $\hat{n}$: $|\psi'\rangle = R(\theta\hat{n})\,|\psi\rangle$.*

Example. *$R\left(\frac{\pi}{2}\hat{y}\right)$ turns $\hat{z}$ into $\hat{x}$, so it takes $|\pm\hat{z}\rangle$ to $|\pm\hat{x}\rangle$ up to phase, and by linearity*

$$R\left(\tfrac{\pi}{2}\hat{y}\right)\left(c_+\,|+\hat{z}\rangle + c_-\,|-\hat{z}\rangle\right) = c_+\,|+\hat{x}\rangle + c_-\,|-\hat{x}\rangle\,.$$

Up to a phase on one column, its S_z matrix is the change-of-basis matrix U of Eq. (37) (exact form in Section 6.4).

Remark 24 (Why the bra carries $R^{\dagger}$)**.** *By the bra-of-a-transformed-state proposition, $\langle\psi'| = \langle\psi|\,R^{\dagger}$. Using $\langle\psi|\,R$ instead fails: with $|+\hat{x}\rangle = R\left(\frac{\pi}{2}\hat{y}\right)|+\hat{z}\rangle$ it would give*

$$\langle+\hat{x}|+\hat{x}\rangle = \langle+\hat{z}|\,R\left(\tfrac{\pi}{2}\hat{y}\right)R\left(\tfrac{\pi}{2}\hat{y}\right)|+\hat{z}\rangle = \langle+\hat{z}|\,R(\pi\hat{y})\,|+\hat{z}\rangle \propto \langle+\hat{z}|-\hat{z}\rangle = 0,$$

since a π rotation about $\hat{y}$ flips $+\hat{z}$ to $-\hat{z}$. That contradicts $\langle+\hat{x}|+\hat{x}\rangle = 1$.

Theorem 2. *Rotation operators are unitary: $R^{\dagger}R = RR^{\dagger} = \mathbb{1}$.*

Proof. A rotation must preserve normalization: $\langle\psi'|\psi'\rangle = \langle\psi|\,R^{\dagger}R\,|\psi\rangle = 1$ for every $|\psi\rangle$, which forces $R^{\dagger}R = \mathbb{1}$. □

6.2 Generators of rotations

Rotations about the same axis add, $R(\theta_1\hat{z})R(\theta_2\hat{z}) = R\big((\theta_1+\theta_2)\hat{z}\big)$, so a finite rotation is $N \gg 1$ small ones,

$$R(\theta\hat{z}) = \left[R\left(\tfrac{\theta}{N}\hat{z}\right)\right]^N. \tag{40}$$

Taylor expanding one small step about the identity and naming the first-order operator $-iJ_z/\hbar$,

$$R\left(\tfrac{\theta}{N}\hat{z}\right) = \mathbb{1} - \frac{i}{\hbar}J_z\frac{\theta}{N} + \mathcal{O}\left(\frac{1}{N^2}\right). \tag{41}$$

Result 13. *By the problem "The exponential as a limit",*

$$R(\theta\hat{z}) = \lim_{N\to\infty}\left(\mathbb{1} - \frac{i}{\hbar}J_z\frac{\theta}{N}\right)^N = e^{-iJ_z\theta/\hbar}, \tag{42}$$

and likewise $R(\theta\hat{n}) = e^{-iJ_n\theta/\hbar}$ for any axis.

Definition 39 (Generator of rotations)**.** *The operator J_z above: the thing that, exponentiated, produces rotations about $\hat{z}$. It has units of $\hbar$, i.e. of angular momentum ($[\hbar] = [\vec{r}\times\vec{p}]$), so $\vec{J}$ is identified as the angular momentum operator. For spin-$\frac{1}{2}$, $\vec{J} = \vec{S}$ and $R(\theta\hat{n}) = e^{-i\theta\hat{S}_n/\hbar}$.*

Proposition 13. *The generator J_z is Hermitian (Def. 36).*

Proof. To first order in θ, $R = \mathbb{1} - i\theta J_z/\hbar$ and $R^\dagger = \mathbb{1} + i\theta J_z^\dagger/\hbar$. Unitarity gives

$$\mathbb{1} = R^\dagger R = \mathbb{1} + \frac{i\theta}{\hbar}\left(J_z^\dagger - J_z\right) + \mathcal{O}(\theta^2),$$

so $J_z^\dagger = J_z$. □

Remark 25. *So J_z qualifies as an observable (Principle 2), consistent with identifying it as angular momentum.*

6.3 Functions of an operator

Proposition 14 (Function of an operator on an eigenstate)**.** *If $\hat{A}\,|a\rangle = a\,|a\rangle$ and f has a Taylor series, then*

$$f(\hat{A})\,|a\rangle = f(a)\,|a\rangle. \tag{43}$$

Proof. $f(\hat{A})$ is defined by its Taylor series, and $\hat{A}^n\,|a\rangle = a^n\,|a\rangle$, so term by term

$$f(\hat{A})\,|a\rangle = \sum_n \frac{f^{(n)}(0)}{n!}\hat{A}^n\,|a\rangle = \sum_n \frac{f^{(n)}(0)}{n!}a^n\,|a\rangle = f(a)\,|a\rangle.$$

□

Proposition 15 (Spectral form)**.** *Writing an operator as a sum of its eigenvalues times the projectors onto its eigenstates is its spectral form. Inserting* $\mathbb{1} = \sum_j |a_j\rangle \langle a_j|$ *and using Proposition 14,*

$$f(\hat{A}) = \sum_j f(a_j) |a_j\rangle \langle a_j| . \tag{44}$$

With $f(x) = x$ *this is the observable operator* $\hat{A} = \sum_j a_j |a_j\rangle \langle a_j|$.

6.3.1 Rotations about $\hat{z}$

Result 14. *With* $f(x) = e^{-i\phi x/\hbar}$ *and* $\hat{S}_z |\pm\hat{z}\rangle = \pm\frac{\hbar}{2} |\pm\hat{z}\rangle$,

$$R(\phi\hat{z}) |\pm\hat{z}\rangle = e^{\mp i\phi/2} |\pm\hat{z}\rangle , \qquad R(\phi\hat{z}) \xrightarrow{S_z} \begin{pmatrix} e^{-i\phi/2} & 0 \\ 0 & e^{i\phi/2} \end{pmatrix} . \tag{45}$$

This makes sense: a rotation about $\hat{z}$ *cannot change* S_z*, so it can only attach phases to the* S_z *states. In other bases the matrix is not diagonal.*

Remark 26. *A full turn,* $\phi = 2\pi$*, gives* $R = -\mathbb{1}$*: a spin-*$\frac{1}{2}$ *state needs* 4π *to return to itself. The sign is a global phase, so it is invisible on its own. Its consequences for many particles are in Section 7.7.*

Problem 10 ($R(\phi\hat{z})$ in the S_x basis)**.** *Find the matrix representation of* $R(\phi\hat{z})$ *in the* S_x *basis.*

6.4 Rotations about any axis

Proposition 16 (Spectral form of a rotation)**.** *For* $\hat{n} \in \{\hat{x}, \hat{y}, \hat{z}\}$,

$$R(\phi\hat{n}) = e^{-i\phi/2} |+\hat{n}\rangle \langle +\hat{n}| + e^{i\phi/2} |-\hat{n}\rangle \langle -\hat{n}| . \tag{46}$$

Proof. Spectral form with $f(x) = e^{-i\phi x/\hbar}$, $\hat{A} = \hat{S}_n$, eigenvalues $\pm\hbar/2$. □

Remark 27 (Recipe)**.** *In any basis* $\{|i\rangle\}$ *this reads* $VDV^\dagger$*, with* $D = \operatorname{diag}\left(e^{-i\phi/2}, e^{i\phi/2}\right)$ *and* V *the matrix whose columns are* $|\pm\hat{n}\rangle$ *written in* $\{|i\rangle\}$*. It is the similarity transformation with the roles of the two bases swapped.*

Proposition 17 ($R(\phi\hat{x})$ in the S_z basis)**.**

$$R(\phi\hat{x}) \xrightarrow{S_z} \begin{pmatrix} \cos\frac{\phi}{2} & -i\sin\frac{\phi}{2} \\ -i\sin\frac{\phi}{2} & \cos\frac{\phi}{2} \end{pmatrix} . \tag{47}$$

Proof. Here $V = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = V^\dagger$, so

$$R(\phi\hat{x}) \xrightarrow{S_z} \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} e^{-i\phi/2} & 0 \\ 0 & e^{i\phi/2} \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} e^{-i\phi/2} + e^{i\phi/2} & e^{-i\phi/2} - e^{i\phi/2} \\ e^{-i\phi/2} - e^{i\phi/2} & e^{-i\phi/2} + e^{i\phi/2} \end{pmatrix} ,$$

and Euler's formula gives $2\cos\frac{\phi}{2}$ and $-2i\sin\frac{\phi}{2}$. □

Remark 28. *This is the same matrix as $R(\phi\hat{z})$ in the S_x basis. It is also the same as $R(\phi\hat{z})$ in the S_y basis (PS2 Q8): with $|\pm\hat{y}\rangle = \frac{1}{\sqrt{2}}(|+\hat{z}\rangle \pm i\,|-\hat{z}\rangle)$,*

$$\langle\pm\hat{y}|\, R(\phi\hat{z})\, |\pm\hat{y}\rangle = \cos\frac{\phi}{2}, \qquad \langle\pm\hat{y}|\, R(\phi\hat{z})\, |\mp\hat{y}\rangle = -i\sin\frac{\phi}{2},$$

so in both bases

$$R(\phi\hat{z}) \longrightarrow \begin{pmatrix} \cos\frac{\phi}{2} & -i\sin\frac{\phi}{2} \\ -i\sin\frac{\phi}{2} & \cos\frac{\phi}{2} \end{pmatrix}.$$

Proposition 18 ($R(\phi\hat{y})$ in the S_z basis)**.**

$$R(\phi\hat{y}) \xrightarrow{S_z} \begin{pmatrix} \cos\frac{\phi}{2} & -\sin\frac{\phi}{2} \\ \sin\frac{\phi}{2} & \cos\frac{\phi}{2} \end{pmatrix}. \tag{48}$$

Proof. Now $V = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ i & -i \end{pmatrix}$ and $V^\dagger = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & -i \\ 1 & i \end{pmatrix} \neq V$. Then

$$R(\phi\hat{y}) \xrightarrow{S_z} \frac{1}{2}\begin{pmatrix} e^{-i\phi/2} + e^{i\phi/2} & -i\left(e^{-i\phi/2} - e^{i\phi/2}\right) \\ i\left(e^{-i\phi/2} - e^{i\phi/2}\right) & e^{-i\phi/2} + e^{i\phi/2} \end{pmatrix},$$

and with $e^{-i\phi/2} - e^{i\phi/2} = -2i\sin\frac{\phi}{2}$ the off-diagonal entries are $\mp\sin\frac{\phi}{2}$. □

Remark 29. *This matrix is real, so $R(\phi\hat{y})$ never introduces factors of i into an S_z column. At $\phi = \pi$ it gives $R(\pi\hat{y})\,|+\hat{z}\rangle = |-\hat{z}\rangle$, the flip used above to show bras carry $R^\dagger$.*

Remark 30 (WARNING: rotation versus change of basis)**.** *Plugging $\phi = \pi/2$ into this proposition does* not *give Bill's change-of-basis matrix:*

$$R\left(\tfrac{\pi}{2}\hat{y}\right) \xrightarrow{S_z} \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \neq U = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

The true rotation sends $|-\hat{z}\rangle$ to $-\,|-\hat{x}\rangle$, not $|-\hat{x}\rangle$, so its second column is $-\,|-\hat{x}\rangle$ while U in Eq. (37) has $+\,|-\hat{x}\rangle$. They differ only by the phase of one basis ket, which is a free choice, so both are valid. For a change of basis with our convention for $|\pm\hat{x}\rangle$, always use U from Eq. (37).

7 Angular Momentum

September 28, 2026

7.1 Non-commuting rotations and commutation relations

Definition 40 (Commutator)**.**

$$[A, B] = AB - BA. \tag{49}$$

A and B commute if $[A, B] = 0$, i.e. the order in which they act does not matter. Directly from the definition, $[B, A] = -[A, B]$ and $[A, BC] = B[A, C] + [A, B]C$.

Rotations about different axes do not commute: $R(\theta\hat{x})R(\theta\hat{y}) \neq R(\theta\hat{y})R(\theta\hat{x})$.

Remark 31. *Rotate a textbook by 90° about $\hat{x}$ then about $\hat{y}$ (right-hand rule). Repeat from the same start with $\hat{y}$ first. The spine ends up pointing in different directions.*

Remark 32 (Why the generators cannot commute)**.** *With $R_x = R(\theta\hat{x}) = e^{-i\theta J_x/\hbar}$ and $R_y = R(\theta\hat{y}) = e^{-i\theta J_y/\hbar}$, one might guess $R_xR_y = e^{-i\theta(J_x+J_y)/\hbar}$. That would force $R_xR_y = R_yR_x$, since $J_x + J_y = J_y + J_x$, which the textbook shows is false. The guess fails because*

$$e^A e^B = e^{A+B} \quad \text{if } [A,B] = 0, \tag{50}$$

but not in general otherwise. So the non-commutativity of rotations tells us $[J_x, J_y] = J_xJ_y - J_yJ_x \neq 0$.

Example ($[S_x, S_y]$ for spin-$\frac{1}{2}$)**.** *With $\hat{S}_j \xrightarrow{S_z} \frac{\hbar}{2}\sigma_j$,*

$$[S_x,S_y] \xrightarrow{S_z} \left(\frac{\hbar}{2}\right)^2(\sigma_x\sigma_y - \sigma_y\sigma_x) = \left(\frac{\hbar}{2}\right)^2\left[\begin{pmatrix} i & 0 \\ 0 & -i\end{pmatrix} - \begin{pmatrix} -i & 0 \\ 0 & i\end{pmatrix}\right] = i\hbar\,\frac{\hbar}{2}\sigma_z \xleftarrow{S_z} i\hbar S_z,$$

so $[S_x, S_y] = i\hbar S_z$.

Result 15 (Angular momentum commutation relations)**.** *The same relation holds for any angular momentum, and cyclic permutations ($x \to y \to z \to x$) give the other two:*

$$[J_x,J_y] = i\hbar J_z, \qquad [J_y,J_z] = i\hbar J_x, \qquad [J_z,J_x] = i\hbar J_y, \tag{51}$$

or compactly $\vec{J}\times\vec{J} = i\hbar\vec{J}$, with $\vec{J} = J_x\hat{x} + J_y\hat{y} + J_z\hat{z}$ and $\times$ the vector cross product (e.g. $\hat{x}\times\hat{y} = \hat{z}$).

7.2 Compatible observables

Definition 41 (Degenerate eigenvalue)**.** *An eigenvalue is degenerate if more than one independent eigenstate has it. Those eigenstates span a* subspace*: a smaller vector space inside the full one, closed under linear combinations.*

Theorem 3. *If $[A, B] = 0$, then A and B have a common eigenbasis.*

Proof. Let $A|a\rangle = a|a\rangle$. Since $AB = BA$,

$$A\big(B|a\rangle\big) = BA|a\rangle = a\big(B|a\rangle\big),$$

so $B|a\rangle$ is also an eigenstate of A with eigenvalue a.

Nondegenerate case. If $|a\rangle$ is the only state with eigenvalue a, then $B|a\rangle = b|a\rangle$, so $|a\rangle$ is an eigenstate of both.

Degenerate case. If n states $|a,i\rangle$ share eigenvalue a, then B maps their subspace into itself, so there it is an $n\times n$ matrix $B_{ij}=\langle a,i|\,B\,|a,j\rangle$. Diagonalizing it (finding the basis in which it is diagonal) gives eigenstates

$$|a,b_j\rangle=\sum_i |a,i\rangle\,\langle a,i|a,b_j\rangle\,,\qquad A\,|a,b_j\rangle=a\,|a,b_j\rangle\,,\qquad B\,|a,b_j\rangle=b_j\,|a,b_j\rangle\,,$$

which are still eigenstates of A because they are combinations of states with the same eigenvalue a. □

Remark 33 (Block-diagonal picture)**.** *Why B cannot mix different eigenvalues of A: if $B\,|a,i\rangle=\sum_{a',j}\langle a',j|B|a,i\rangle\,|a',j\rangle$ contained states with $a'\neq a$, then*

$$A\big(B\,|a,i\rangle\,\big)=\sum_{a',j}\langle a',j|B|a,i\rangle\;a'\,|a',j\rangle\,,$$

which equals $a\big(B\,|a,i\rangle\,\big)$ only if $\langle a',j|B|a,i\rangle\propto\delta_{aa'}$. So in the basis $\{|a_i,j\rangle\}$, B is block diagonal, with one block per eigenvalue of A:

$$B\xrightarrow{\{|a_i,j\rangle\}}\begin{pmatrix}B(a_1) & 0 & \cdots\\ 0 & B(a_2) & \\ \vdots & & \ddots\end{pmatrix},\qquad B_{kl}(a_i)=\langle a_i,k|B|a_i,l\rangle\,.$$

Diagonalizing each block separately gives the simultaneous eigenstates, $|a,b_j\rangle=\sum_i\langle a,i|a,b_j\rangle\,|a,i\rangle$.

Definition 42 (Compatible observables)**.** *Observables that commute, and so can have definite values at the same time. Otherwise they are incompatible.*

Remark 34. *$[J_x,J_y]=i\hbar J_z\neq 0$, so the components of angular momentum are incompatible: they have no common eigenbasis (for spin-$\frac{1}{2}$, no common eigenstate at all). This is the operator version of Experiment 3: putting $|+\hat z\rangle$ into an SGx gives a 50/50 mix of $|\pm\hat x\rangle$, so no state gives $S_z=+\hbar/2$ and $S_x=+\hbar/2$ with certainty. Preparing an eigenstate of one observable leaves the outcomes of an incompatible one uncertain; for $|+\hat z\rangle$ and S_x the uncertainty is maximal, $\Delta S_x=\hbar/2$.*

7.3 Uncertainty relations

Result 16 (Generalized uncertainty relation)**.** *For observables A and B, in any state,*

$$\Delta A\,\Delta B\geq\frac{1}{2}\,|\langle[A,B]\rangle|\,. \tag{52}$$

The derivation is in Townsend, Eq. (3.74). The bound only says something when $[A,B]\neq 0$.

Result 17. *For angular momentum, $[J_x,J_y]=i\hbar J_z$ gives*

$$\Delta J_x\,\Delta J_y\geq\frac{\hbar}{2}\,|\langle J_z\rangle|\,. \tag{53}$$

Example. *For $|+\hat{z}\rangle$, $\langle S_z\rangle = \hbar/2$, and both S_x and S_y give $\pm\hbar/2$ with probability $\frac{1}{2}$, so $\Delta S_x = \Delta S_y = \hbar/2$. Then $\Delta S_x \, \Delta S_y = \hbar^2/4 = \frac{\hbar}{2}\langle S_z\rangle$: the bound is saturated.*

Remark 35. *Position and momentum will also turn out not to commute, $[x, p_x] = i\hbar$, which gives the Heisenberg uncertainty relation*

$$\Delta x \, \Delta p_x \geq \frac{\hbar}{2}. \tag{54}$$

7.4 Total angular momentum

Definition 43 (Total angular momentum)**.** *The generator $\vec{J}$ of Section 6.2 is the total angular momentum, the sum of the orbital and spin angular momenta:*

$$\vec{J} = \vec{L} + \vec{S}. \tag{55}$$

Definition 44 (Square of the angular momentum)**.**

$$\vec{J}^2 = J_x^2 + J_y^2 + J_z^2. \tag{56}$$

Proposition 19. *$\vec{J}^2$ commutes with every component of $\vec{J}$; in particular $[\vec{J}^2, J_z] = 0$.*

Proof. Using $[A, BC] = B[A, C] + [A, B]C$:

$$\begin{aligned}
[J_z, J_x^2] &= J_x[J_z, J_x] + [J_z, J_x]J_x = i\hbar\,(J_xJ_y + J_yJ_x)\,, \\
[J_z, J_y^2] &= J_y[J_z, J_y] + [J_z, J_y]J_y = -i\hbar\,(J_yJ_x + J_xJ_y)\,, \\
[J_z, J_z^2] &= 0,
\end{aligned}$$

using $[J_z, J_y] = -i\hbar J_x$. The first two cancel, so $[J_z, \vec{J}^2] = 0$. The same steps work for x and y. □

Remark 36. *$\vec{J}^2$ and J_z are compatible, so they share an eigenbasis.*

Definition 45 (Angular momentum eigenstates)**.** *The simultaneous eigenstates of $\vec{J}^2$ and J_z are labelled $|j, m\rangle$, with $\langle j, m|j, m'\rangle = \delta_{mm'}$. By convention their eigenvalues are written*

$$\vec{J}^2\,|j, m\rangle = \hbar^2 j(j+1)\,|j, m\rangle\,, \qquad J_z\,|j, m\rangle = \hbar m\,|j, m\rangle\,. \tag{57}$$

For spin, $j = s$, matching the labels $|s, m\rangle$ and the magnitude $|\vec{S}| = \hbar\sqrt{s(s+1)}$ used earlier. The allowed values of j and m are derived in Section 7.5.

7.5 Ladder operators

Definition 46 (Ladder operators)**.**

$$J_\pm = J_x \pm iJ_y, \qquad J_x = \frac{1}{2}\left(J_+ + J_-\right), \qquad J_y = \frac{1}{2i}\left(J_+ - J_-\right). \tag{58}$$

They are not Hermitian: $J_+^\dagger = J_x - iJ_y = J_-$.

Proposition 20.

$$[J_z, J_\pm] = \pm\hbar J_\pm. \tag{59}$$

Proof. $[J_z, J_\pm] = [J_z, J_x] \pm i[J_z, J_y] = i\hbar J_y \pm i(-i\hbar J_x) = \pm\hbar\left(J_x \pm iJ_y\right)$. □

Proposition 21. $J_\pm \left|j,m\right\rangle$ *is an eigenstate of* J_z *with eigenvalue* $\hbar(m \pm 1)$*, so* $J_\pm \left|j,m\right\rangle \propto \left|j,m\pm1\right\rangle$.

Proof. From $[J_z, J_-] = -\hbar J_-$, $J_zJ_- = J_-J_z - \hbar J_- = J_-(J_z - \hbar)$, so

$$J_z\big(J_-\left|j,m\right\rangle\big) = J_-(J_z - \hbar)\left|j,m\right\rangle = \hbar(m-1)\big(J_-\left|j,m\right\rangle\big).$$

The same steps with $J_zJ_+ = J_+(J_z+\hbar)$ give $\hbar(m+1)$. Since $\vec{J}^2$ commutes with J_x and J_y, it commutes with $J_\pm$, so j is unchanged. □

Proposition 22 (Action of the ladder operators)**.**

$$J_\pm \left|j,m\right\rangle = \hbar\sqrt{j(j+1) - m(m\pm1)}\,\left|j,m\pm1\right\rangle. \tag{60}$$

J_- *is the* lowering operator *and* J_+ *the* raising operator.

Proof. Since $J_+ = J_-^\dagger$, the squared norm of $J_-\left|j,m\right\rangle$ is $\langle j,m|J_+J_-|j,m\rangle$. Expanding,

$$J_+J_- = (J_x + iJ_y)(J_x - iJ_y) = J_x^2 + J_y^2 - i[J_x, J_y] = \vec{J}^2 - J_z^2 + \hbar J_z,$$

so with Eq. (57),

$$\langle j,m|J_+J_-|j,m\rangle = \hbar^2\left[j(j+1) - m^2 + m\right] = \hbar^2\left[j(j+1) - m(m-1)\right].$$

Taking the coefficient real and positive (a phase convention) gives the J_- result. The same steps with $J_-J_+ = \vec{J}^2 - J_z^2 - \hbar J_z$ give the J_+ result. □

Result 18 (Allowed values of j and m)**.** *The ladder stops at both ends:*

$$J_+\left|j,j\right\rangle = \hbar\sqrt{j(j+1) - j(j+1)}\,\left|j,j+1\right\rangle = 0, \qquad J_-\left|j,-j\right\rangle = \hbar\sqrt{j(j+1) - j(j+1)}\,\left|j,-j-1\right\rangle = 0.$$

Starting from $m = j$ *and lowering,* $J_-\left|j,j\right\rangle \propto \left|j,j-1\right\rangle$, $J_-^2\left|j,j\right\rangle \propto \left|j,j-2\right\rangle$, $\ldots$, $J_-^{2j}\left|j,j\right\rangle \propto \left|j,-j\right\rangle$. *Reaching* $-j$ *in whole steps requires* $2j$ *to be a whole number, so*

$$j = 0, \tfrac{1}{2}, 1, \tfrac{3}{2}, \ldots, \qquad m = -j, -j+1, \ldots, j-1, j, \tag{61}$$

giving $2j+1$ *states for each* j*, as in Eq.* (15).

Remark 37. *The ladder cannot run past these ends because squared norms are non-negative:* $j(j+1) - m(m\pm1) \geq 0$ *keeps* $-j \leq m \leq j$.

7.6 Matrix elements

Result 19. *In the* $\{|j,m\rangle\}$ *basis,*

$$\langle j,m'|J_\pm|j,m\rangle = \hbar\sqrt{j(j+1)-m(m\pm1)}\,\delta_{m',m\pm1}, \qquad \langle j,m'|J_z|j,m\rangle = \hbar m\,\delta_{m'm}. \tag{62}$$

Example (Spin 1)**.** *For* $j=1$ *there are three states,* $m=1,0,-1$ *(e.g. the deuteron). Ordering the basis as* $\{|1,1\rangle, |1,0\rangle, |1,-1\rangle\}$*,*

$$J_z \longrightarrow \hbar\begin{pmatrix}1&0&0\\0&0&0\\0&0&-1\end{pmatrix}, \qquad J_+ \longrightarrow \hbar\begin{pmatrix}0&\sqrt{2}&0\\0&0&\sqrt{2}\\0&0&0\end{pmatrix}, \qquad J_- \longrightarrow \hbar\begin{pmatrix}0&0&0\\\sqrt{2}&0&0\\0&\sqrt{2}&0\end{pmatrix}.$$

With $J_x=\frac{1}{2}(J_+ + J_-)$ *and* $J_y=\frac{1}{2i}(J_+ - J_-)$*,*

$$J_x \longrightarrow \frac{\hbar}{\sqrt{2}}\begin{pmatrix}0&1&0\\1&0&1\\0&1&0\end{pmatrix}, \qquad J_y \longrightarrow \frac{i\hbar}{\sqrt{2}}\begin{pmatrix}0&-1&0\\1&0&-1\\0&1&0\end{pmatrix}.$$

Check $[J_x,J_y]=i\hbar J_z$*:*

$$[J_x,J_y] \longrightarrow \frac{i\hbar^2}{2}\left[\begin{pmatrix}1&0&-1\\0&0&0\\1&0&-1\end{pmatrix}-\begin{pmatrix}-1&0&-1\\0&0&0\\1&0&1\end{pmatrix}\right] = i\hbar\cdot\hbar\begin{pmatrix}1&0&0\\0&0&0\\0&0&-1\end{pmatrix} \longleftarrow i\hbar J_z.$$

7.7 Rotations by 2π: fermions and bosons

Using the spectral form (Section 6.4) with j fixed,

$$R(2\pi\hat{z}) = e^{-2\pi i J_z/\hbar}\sum_m |j,m\rangle\langle j,m| = \sum_m e^{-2\pi i m}\,|j,m\rangle\langle j,m|\,.$$

Result 20. $R(2\pi\hat{z})=\mathbb{1}$ *only if every* m *is an integer, i.e.* $j=0,1,2,\ldots$*. For half-integer* j*, every* m *is a half-integer and* $R(2\pi\hat{z})=-\mathbb{1}$*.*

Remark 38. *If we demanded that a* 2π *rotation leave every state unchanged,* j *would have to be an integer. But electrons have* $j=\frac{1}{2}$*, with* $m=\pm\frac{1}{2}$ *and* $S_z=\pm\hbar/2$*, so* $R(2\pi\hat{z})=e^{\mp i\pi}=-\mathbb{1}$ *(Remark 26). For a single spin-*$\frac{1}{2}$ *particle this is a global phase,* $R(2\pi\hat{z})\,|\psi\rangle = -\,|\psi\rangle = e^{i\pi}\,|\psi\rangle$*, with no effect on observables. For several particles, this sign for half-integer spin results in the Pauli exclusion principle.*

Definition 47 (Fermions and bosons)**.** *Particles with half-integer spin* $j=\frac{1}{2},\frac{3}{2},\frac{5}{2},\ldots$ *are* fermions *and obey Pauli exclusion. Particles with integer spin* $j=0,1,2,\ldots$ *are* bosons*.*

Remark 39. *Loosely, this connection to rotations is related to the* exchange *of two identical particles: swapping them is like rotating one around the other.*

7.8 Atomic quantum numbers

Definition 48 (Atomic quantum numbers)**.** *An electron in an atom is labelled* $|n, l, m_l, s, m_s\rangle$*, where*

- $n = 1, 2, 3, \ldots$ *is the* principal *quantum number: the size and energy of the electron cloud.*
- $l = 0, 1, \ldots, n-1$ *is the* azimuthal *quantum number: the shape of the electron cloud, with* $|\vec{L}| = \hbar\sqrt{l(l+1)}$*. The values* $l = 0, 1, 2, 3$ *are called s, p, d, f.*
- $m_l = -l, -l+1, \ldots, +l$ *is the* magnetic *quantum number: the spatial orientation of the electron cloud, with* $L_z = m_l\hbar$*.*
- $s = \frac{1}{2}$ *is the* spin *quantum number: the intrinsic angular momentum of the electron, with* $m_s = \pm\frac{1}{2}$ *giving* $S_z = m_s\hbar$*.*

The total angular momentum $\vec{J} = \vec{L} + \vec{S}$ *has its own quantum number* j*, running from* $|l - s|$ *to* $l + s$ *in steps of 1. For an electron,* $j = l \pm \frac{1}{2}$ *(only* $j = \frac{1}{2}$ *when* $l = 0$*).*

Remark 40. *No two electrons in an atom share all of* n*,* l*,* m_l*,* m_s *(Pauli exclusion principle). A subshell* l *has* $2l + 1$ *values of* m_l*, each with two values of* m_s*, so it holds* $2(2l + 1)$ *electrons: 2 for s, 6 for p, 10 for d, 14 for f.*

Remark 41. *Two notation clashes: the subshell name s (*$l = 0$*) is not the spin quantum number s, and the quantum number* j *is not the outcome label* j *in* $|j\rangle$*.*

8 Time Evolution

September 30 - October 16, 2026

8.1 The time-evolution operator

A small rotation is generated by angular momentum (Section 6.2),

$$R(d\phi\,\hat{n}) = \mathbb{1} - \frac{i}{\hbar}\left(\vec{J}\cdot\hat{n}\right)d\phi, \qquad R(\phi\hat{n})\,|\psi_0\rangle = |\psi_\phi\rangle\,,$$

where ϕ is the parameter that labels how far the state has been rotated. Time evolution works the same way, with t as the parameter.

Definition 49 (Time-evolution operator)**.** *The linear operator* $U(t)$ *that takes a state from time* 0 *to time* t*:*

$$|\psi(t)\rangle = U(t)\,|\psi(0)\rangle\,. \tag{63}$$

Remark 42. *Not to be confused with the change-of-basis matrix* U *of Section 5.4. Both are unitary, which is where the shared letter comes from.*

Proposition 23. $U(t)$ *is unitary:* $U^\dagger(t)U(t) = \mathbb{1}$*.*

Proof. Probabilities must still add to 1 at every time,

$$\langle\psi(t)|\psi(t)\rangle = \langle\psi(0)|\, U^\dagger(t)U(t)\,|\psi(0)\rangle = \langle\psi(0)|\psi(0)\rangle = 1$$

for every $|\psi(0)\rangle$, which forces $U^\dagger U = \mathbb{1}$. □

Definition 50 (Hamiltonian)**.** *The generator of time evolution, defined by the first-order expansion*

$$U(dt) = \mathbb{1} - \frac{i}{\hbar} H\, dt. \tag{64}$$

It has units $[H] = [\hbar/t]$ = energy, so H is identified as the energy operator.

Remark 43. *The same argument as for J_z (Section 6.2) shows that unitarity of U forces $H = H^\dagger$, so energy is an observable (Principle 2).*

Proposition 24 (Schrödinger equation)**.**

$$\frac{d}{dt}\,|\psi(t)\rangle = -\frac{i}{\hbar} H\,|\psi(t)\rangle\,. \tag{65}$$

Proof. Two evolutions in a row compose,

$$U(t+\Delta t) = U(\Delta t)U(t) = \left(\mathbb{1} - \frac{i}{\hbar}H\Delta t + \mathcal{O}(\Delta t^2)\right) U(t),$$

so

$$\frac{dU}{dt} = \lim_{\Delta t\to 0} \frac{U(t+\Delta t) - U(t)}{\Delta t} = -\frac{i}{\hbar} HU(t).$$

Acting on $|\psi(0)\rangle$ gives the result. □

Result 21 (Time-independent Hamiltonian)**.** *If H does not depend on time,*

$$U(t) = e^{-iHt/\hbar}. \tag{66}$$

Check: $\frac{d}{dt}e^{-iHt/\hbar} = -\frac{i}{\hbar}He^{-iHt/\hbar}$, and the initial condition $U(0) = e^0 = \mathbb{1}$ holds.

8.2 Stationary states

Definition 51 (Stationary state)**.** *A state that does not evolve into a physically different state. These are the eigenstates of H, the energy eigenstates $H\,|E\rangle = E\,|E\rangle$.*

Proposition 25.

$$U(t)\,|E\rangle = e^{-iEt/\hbar}\,|E\rangle\,. \tag{67}$$

This is a global phase, so $|E\rangle$ and $U(t)\,|E\rangle$ are the same physical state for all t.

Proof. Proposition 14 with $f(x) = e^{-ixt/\hbar}$. □

Result 22 (Evolution of a general state)**.** *Inserting the identity in the energy eigenbasis,* $|\psi(0)\rangle = \sum_n |E_n\rangle \langle E_n|\psi(0)\rangle$*, and evolving each term,*

$$|\psi(t)\rangle = \sum_n \langle E_n|\psi(0)\rangle \; e^{-iE_n t/\hbar} \, |E_n\rangle \, . \tag{68}$$

Recipe: expand in energy eigenstates, then attach $e^{-iE_n t/\hbar}$ *to each. The phases are different for each term, so they are relative phases and the state changes physically unless only one* E_n *appears.*

8.3 Time dependence of expectation values

Proposition 26 (Equation of motion)**.**

$$\frac{d}{dt}\langle A\rangle = \frac{i}{\hbar}\langle [H,A]\rangle + \langle \frac{\partial A}{\partial t}\rangle \, . \tag{69}$$

Proof. With $\langle A\rangle = \langle\psi(t)| \, A \, |\psi(t)\rangle$ and the chain rule,

$$\frac{d}{dt}\langle A\rangle = \left(\frac{d}{dt}\langle\psi(t)|\right) A \, |\psi(t)\rangle + \langle\psi(t)| \, A \left(\frac{d}{dt}|\psi(t)\rangle\right) + \langle\psi(t)| \, \frac{\partial A}{\partial t} \, |\psi(t)\rangle \, .$$

The Schrödinger equation and its bra (using $H^\dagger = H$) give $\frac{d}{dt}|\psi\rangle = -\frac{i}{\hbar}H|\psi\rangle$ and $\frac{d}{dt}\langle\psi| = \frac{i}{\hbar}\langle\psi|H$, so the first two terms are $\frac{i}{\hbar}\langle\psi|\,(HA - AH)\,|\psi\rangle$. □

Remark 44. *The* $\partial A/\partial t$ *term is only there if the operator itself has explicit time dependence, e.g. a time-dependent field or external force (Section 8.5).*

Definition 52 (Constant of the motion)**.** *An observable* A *with* $\partial A/\partial t = 0$ *and* $[H,A] = 0$*. Then* $\frac{d}{dt}\langle A\rangle = 0$*: it is a conserved quantity.*

Example. $A = H$ *always commutes with itself, so energy is conserved whenever* $\partial H/\partial t = 0$*.*

8.4 Spin precession

Put an electron ($q = -|e|$) in a uniform field $\vec{B} = B_0\hat{z}$. With $\vec{\mu} = -g\mu_B\vec{S}/\hbar$,

$$H_0 = -\vec{\mu}\cdot\vec{B} = \omega_0 S_z, \qquad \omega_0 = \frac{g\mu_B B_0}{\hbar}, \qquad [\omega_0] = \frac{[\text{energy}]}{[\hbar]} = [\text{frequency}]. \tag{70}$$

Since $[H_0, S_z] = 0$, S_z is a constant of the motion. The energy eigenstates are

$$H_0\,|\pm\hat{z}\rangle = \pm\frac{\hbar\omega_0}{2}\,|\pm\hat{z}\rangle \, .$$

Result 23 (Time evolution is a rotation)**.**

$$U(t) = e^{-i\omega_0 t S_z/\hbar} = R(\phi\hat{z})\Big|_{\phi=\omega_0 t}, \qquad U(t)\,|\pm\hat{z}\rangle = e^{\mp i\omega_0 t/2}\,|\pm\hat{z}\rangle \, . \tag{71}$$

Evolving under $H_0 = \omega_0 S_z$ *is the same as rotating about* $\hat{z}$ *by* $\phi = \omega_0 t$*. The states* $|\pm\hat{z}\rangle$ *are stationary, equivalently invariant under rotations about* $\hat{z}$*.*

Example (Starting in $|+\hat{x}\rangle$)**.** *$|+\hat{x}\rangle$ is a superposition of energy eigenstates, so by Eq.* (68)

$$|\psi(t)\rangle = e^{-iH_0 t/\hbar}\frac{1}{\sqrt{2}}\left(|+\hat{z}\rangle + |-\hat{z}\rangle\right) = \frac{1}{\sqrt{2}}\left(e^{-i\omega_0 t/2}\,|+\hat{z}\rangle + e^{i\omega_0 t/2}\,|-\hat{z}\rangle\right).$$

With $\langle +\hat{x}| \pm \hat{z}\rangle = \frac{1}{\sqrt{2}}$,

$$P_{+\hat{x}}(t) = |\langle +\hat{x}|\psi(t)\rangle|^2 = \left|\frac{1}{2}e^{-i\omega_0 t/2} + \frac{1}{2}e^{i\omega_0 t/2}\right|^2 = \cos^2\frac{\omega_0 t}{2} = \frac{1}{2}\left(1+\cos\omega_0 t\right). \tag{72}$$

8.4.1 Expectation values

With $S_x \xrightarrow{S_z} \frac{\hbar}{2}\sigma_x$ and $\psi(t) \xrightarrow{S_z} \frac{1}{\sqrt{2}}\left(e^{-i\omega_0 t/2}, e^{i\omega_0 t/2}\right)^T$,

$$\langle S_x(t)\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} e^{i\omega_0 t/2} & e^{-i\omega_0 t/2}\end{pmatrix}\frac{\hbar}{2}\begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}\frac{1}{\sqrt{2}}\begin{pmatrix}e^{-i\omega_0 t/2}\\ e^{i\omega_0 t/2}\end{pmatrix}$$

$$= \frac{\hbar}{4}\left(e^{i\omega_0 t} + e^{-i\omega_0 t}\right) = \frac{\hbar}{2}\cos\omega_0 t.$$

Alternatively, from the probabilities,

$$\langle S_x(t)\rangle = \frac{\hbar}{2}\left(P_{+\hat{x}} - P_{-\hat{x}}\right) = \frac{\hbar}{2}\left(2P_{+\hat{x}} - 1\right) = \frac{\hbar}{2}\cos\omega_0 t.$$

The same matrix product with $S_y \xrightarrow{S_z} \frac{\hbar}{2}\sigma_y$ gives $\langle S_y(t)\rangle = \frac{\hbar}{2}\sin\omega_0 t$.

Result 24 (Precession)**.**

$$\langle \vec{S}(t)\rangle = \frac{\hbar}{2}\left(\cos\omega_0 t\,\hat{x} + \sin\omega_0 t\,\hat{y}\right), \qquad \langle S_z\rangle = 0. \tag{73}$$

The average spin rotates counterclockwise about $\hat{z}$ at angular frequency ω_0 (Larmor precession), while $\langle S_z\rangle$ stays constant as a constant of the motion should.

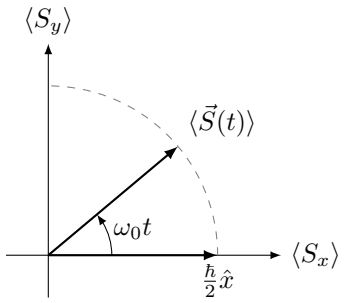


Figure 3: Precession of $\langle \vec{S}\rangle$ in the x-y plane, starting from $|+\hat{x}\rangle$.

8.4.2 From the equation of motion

The same result follows without ever writing the state. Using $[S_z, S_x] = i\hbar S_y$ and $[S_z, S_y] = -i\hbar S_x$,

$$\frac{d}{dt}\langle S_x\rangle = \frac{i}{\hbar}\langle[\omega_0 S_z, S_x]\rangle = -\omega_0\langle S_y\rangle, \tag{74}$$

$$\frac{d}{dt}\langle S_y\rangle = \frac{i}{\hbar}\langle[\omega_0 S_z, S_y]\rangle = \omega_0\langle S_x\rangle. \tag{75}$$

Differentiating Eq. (74) and using Eq. (75),

$$\frac{d^2}{dt^2}\langle S_x\rangle = -\omega_0\frac{d}{dt}\langle S_y\rangle = -\omega_0^2\langle S_x\rangle,$$

with general solution $\langle S_x\rangle = A\cos\omega_0 t + B\sin\omega_0 t$. The initial conditions are $\langle S_x(0)\rangle = \hbar/2$ and, from Eq. (74), $\frac{d}{dt}\langle S_x\rangle\big|_{t=0} = -\omega_0\langle S_y(0)\rangle = 0$, so $A = \hbar/2$, $B = 0$. Then $\langle S_y\rangle = -\frac{1}{\omega_0}\frac{d}{dt}\langle S_x\rangle$ gives

$$\langle S_x(t)\rangle = \frac{\hbar}{2}\cos\omega_0 t, \qquad \langle S_y(t)\rangle = \frac{\hbar}{2}\sin\omega_0 t.$$

8.5 Magnetic resonance

Definition 53 (Magnetic resonance)**.** *Driving transitions between spin states with an oscillating field tuned to the precession frequency. Applications:*

- Nuclear magnetic resonance *(NMR), on nuclear spin $\vec{I}$: e.g. medical imaging (MRI), which maps local water content through resonant absorption of radio-frequency (∼MHz) radiation.*
- Electron spin resonance *(ESR), on electron spin $\vec{S}$: e.g. characterizing materials with microwave (∼GHz) radiation.*
- Qubit control*: single-qubit gates (unitary operations) on qubits made from electron spins, nuclear spins, ions, neutral atoms or superconducting circuits.*

The field is a strong longitudinal part plus a weaker oscillating transverse part,

$$\vec{B}(t) = B_0\hat{z} + B_1\cos\omega t\,\hat{x}, \qquad B_0 \gg B_1, \tag{76}$$

so the Hamiltonian now has explicit time dependence:

$$H(t) = -\vec{\mu}\cdot\vec{B}(t) = \omega_0 S_z + \Omega\cos(\omega t)\,S_x, \tag{77}$$

with, using the magnetons defined with the Stern-Gerlach experiment,

$$\text{ESR } (q = -|e|): \quad \omega_0 = \frac{g\mu_B B_0}{\hbar}, \qquad \Omega = \frac{g\mu_B B_1}{\hbar},$$

$$\text{NMR } (q = +|e|): \quad \omega_0 = -\frac{g_I\mu_N B_0}{\hbar}, \qquad \Omega = -\frac{g_I\mu_N B_1}{\hbar}.$$

For $B_0 \sim 1$ T, $\omega_0/2\pi$ is in the GHz range for ESR ($g \approx 2$) and the MHz range for NMR, since $\mu_N \approx \mu_B/1836$.

Remark 45 (Angular vs linear frequency)**.** *ω_0 is an angular frequency, in rad/s. $\omega_0/2\pi$ is the linear frequency, in Hz = cycles/s, with 1 cycle = 2π rad.*

Remark 46. *With H time-dependent, $U(t) = e^{-iHt/\hbar}$ no longer holds, and we must work with the Schrödinger equation directly, $\partial_t |\psi(t)\rangle = -\frac{i}{\hbar} H(t) |\psi(t)\rangle$.*

8.5.1 The rotating frame

Split the Hamiltonian into a large static part and a small driving part,

$$H(t) = H_0 + V(t), \qquad H_0 = \omega_0 S_z, \qquad V(t) = \Omega \cos(\omega t)\, S_x.$$

Since $\omega_0 \gg \Omega$, most of the evolution is the fast precession from H_0. Factor it out:

$$|\psi(t)\rangle = U_0(t)\, |\tilde{\psi}(t)\rangle\,, \qquad U_0(t) = e^{-iH_0 t/\hbar}, \tag{78}$$

where U_0 is the "fast" part and $|\tilde{\psi}(t)\rangle$ the "slow" part.

Proposition 27 (Schrödinger equation in the rotating frame)**.**

$$\partial_t |\tilde{\psi}(t)\rangle = -\frac{i}{\hbar} \tilde{V}(t)\, |\tilde{\psi}(t)\rangle\,, \qquad \tilde{V}(t) = U_0^\dagger(t) V(t) U_0(t). \tag{79}$$

Proof. Substituting into the Schrödinger equation with $\dot{U}_0 = -\frac{i}{\hbar} H_0 U_0$,

$$-\frac{i}{\hbar} H_0 U_0\, |\tilde{\psi}\rangle + U_0\, \partial_t |\tilde{\psi}\rangle = -\frac{i}{\hbar} (H_0 + V)\, U_0\, |\tilde{\psi}\rangle\,.$$

The H_0 terms cancel. Multiplying on the left by $U_0^\dagger$ gives the result. □

Definition 54 (Rotating frame)**.** *Since $U_0(t) = e^{-i\omega_0 t S_z/\hbar} = R(\omega_0 t\, \hat{z})$ is a rotation about $\hat{z}$ by the time-dependent angle $\omega_0 t$, describing the system by $|\tilde{\psi}(t)\rangle$ is called going into the rotating frame.*

To compute $\tilde{V}$, write $S_x = \frac{1}{2}(S_+ + S_-)$ with the spin-$\frac{1}{2}$ ladder operators (Section 7.5),

$$S_+ = \hbar\, |+\hat{z}\rangle \langle -\hat{z}|\,, \qquad S_- = \hbar\, |-\hat{z}\rangle \langle +\hat{z}|\,.$$

Lemma 4.

$$e^{i\omega_0 t S_z/\hbar}\, S_\pm\, e^{-i\omega_0 t S_z/\hbar} = e^{\pm i\omega_0 t} S_\pm. \tag{80}$$

Proof. For S_+: $e^{-i\omega_0 t S_z/\hbar}$ acting right on $|-\hat{z}\rangle$ gives $e^{i\omega_0 t/2}$, and $e^{i\omega_0 t S_z/\hbar}$ acting left on $|+\hat{z}\rangle$ gives $e^{i\omega_0 t/2}$, a total of $e^{i\omega_0 t}$. Swapping the roles gives $e^{-i\omega_0 t}$ for S_-. □

With $\cos\omega t = \frac{1}{2}\left(e^{i\omega t} + e^{-i\omega t}\right)$,

$$\begin{aligned} \tilde{V}(t) &= \frac{\Omega}{4} \left(e^{i\omega t} + e^{-i\omega t}\right) \left(e^{i\omega_0 t} S_+ + e^{-i\omega_0 t} S_-\right) \\ &= \frac{\Omega}{4} \Big(\underbrace{e^{-i(\omega-\omega_0)t} S_+ + e^{i(\omega-\omega_0)t} S_-}_{\text{slow for } \omega \approx \omega_0} + \underbrace{e^{i(\omega+\omega_0)t} S_+ + e^{-i(\omega+\omega_0)t} S_-}_{\text{fast}} \Big). \end{aligned} \tag{81}$$

8.5.2 Rotating-wave approximation

Integrating the rotating-frame equation,

$$|\tilde{\psi}(t)\rangle - |\tilde{\psi}(0)\rangle = -\frac{i}{\hbar}\int_0^t dt'\,\tilde{V}(t')\,|\tilde{\psi}(t')\rangle\,.$$

The fast terms $\sim e^{\pm i(\omega+\omega_0)t}$ oscillate many times while $|\tilde{\psi}\rangle$ barely changes, so they average to zero on a time scale $t \gtrsim 1/(\omega+\omega_0)$. Near resonance ($\omega \approx \omega_0$) the slow terms $\sim e^{\pm i(\omega-\omega_0)t}$ dominate.

Definition 55 (Rotating-wave approximation (RWA))**.** *Dropping the terms* $\sim e^{\pm i(\omega+\omega_0)t}$*. The corrections are small when* $\Omega/\omega_0 \ll 1$*, i.e.* $B_0 \gg B_1$*. Then*

$$\tilde{V}(t) \simeq \frac{\Omega}{4}\left(e^{-i\delta t}S_+ + e^{i\delta t}S_-\right). \tag{82}$$

Definition 56 (Detuning)**.** $\delta = \omega - \omega_0$*, how far the drive is from resonance.*

Result 25 (On resonance)**.** *For* $\omega = \omega_0$*,* $\delta = 0$ *and* $\tilde{V}$ *is time-independent:*

$$\tilde{V} \simeq \frac{\Omega}{4}(S_+ + S_-) = \frac{\Omega}{2}S_x, \qquad |\tilde{\psi}(t)\rangle \simeq e^{-i\Omega t S_x/2\hbar}\,|\psi(0)\rangle = R\left(\tfrac{\Omega t}{2}\hat{x}\right)|\psi(0)\rangle\,, \tag{83}$$

using $|\tilde{\psi}(0)\rangle = |\psi(0)\rangle$ *since* $U_0(0) = \mathbb{1}$*. In the rotating frame the spin simply rotates about* $\hat{x}$ *at angular frequency* $\Omega/2$*.*

8.5.3 Rabi oscillations

Start in $|\psi(0)\rangle = |+\hat{z}\rangle = \frac{1}{\sqrt{2}}(|+\hat{x}\rangle + |-\hat{x}\rangle)$, i.e. expand in the eigenstates of $\tilde{V} \propto S_x$ as in Eq. (68). With $S_x\,|\pm\hat{x}\rangle = \pm\frac{\hbar}{2}\,|\pm\hat{x}\rangle$,

$$|\tilde{\psi}(t)\rangle \simeq \frac{1}{\sqrt{2}}\left(e^{-i\Omega t/4}\,|+\hat{x}\rangle + e^{i\Omega t/4}\,|-\hat{x}\rangle\right)$$

$$|\tilde{\psi}(t)\rangle \xrightarrow{S_x} \frac{1}{\sqrt{2}}\begin{pmatrix} e^{-i\Omega t/4} \\ e^{i\Omega t/4}\end{pmatrix}.$$

Since U_0 is a function of S_z, $[U_0, S_z] = 0$, so

$$\langle S_z\rangle = \langle\psi(t)|\,S_z\,|\psi(t)\rangle = \langle\tilde{\psi}(t)|\,S_z\,|\tilde{\psi}(t)\rangle\,.$$

With $S_z \xrightarrow{S_x} \frac{\hbar}{2}\sigma_x$ (problem "$\hat{S}_z$ in the S_x basis"),

$$\langle S_z\rangle = \frac{1}{2}\begin{pmatrix} e^{i\Omega t/4} & e^{-i\Omega t/4}\end{pmatrix}\frac{\hbar}{2}\begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}\begin{pmatrix} e^{-i\Omega t/4} \\ e^{i\Omega t/4}\end{pmatrix} = \frac{\hbar}{4}\left(e^{i\Omega t/2} + e^{-i\Omega t/2}\right).$$

Result 26 (Rabi oscillations)**.**

$$\langle S_z(t)\rangle = \frac{\hbar}{2}\cos\frac{\Omega t}{2}, \qquad P_{+\hat{z}}(t) = \cos^2\frac{\Omega t}{4}. \tag{84}$$

The spin flips back and forth between $|+\hat{z}\rangle$ *and* $|-\hat{z}\rangle$*. A drive left on for* $\Omega t/2 = \pi$ *(a "π-pulse") takes* $|+\hat{z}\rangle$ *to* $|-\hat{z}\rangle$*: this is how a spin qubit is flipped.*

Remark 47 (WARNING: factor of 2)**.** *Bill's notes write the phases as* $e^{\mp i\Omega t/2}$ *and get* $\langle S_z\rangle = \frac{\hbar}{2}\cos\Omega t$*. The eigenvalues of* S_x *are* $\pm\hbar/2$*, not* $\pm\hbar$*, so the phases are* $e^{\mp i\Omega t/4}$ *and the Rabi frequency is* $\Omega/2$*. A numerical solution of the full* $H(t)$ *confirms* $\cos(\Omega t/2)$*. The half comes from the RWA:* $\cos\omega t$ *splits into two counter-rotating pieces and only one of them is kept.*

A Solutions

Solution 1 (The exponential as a limit)**.** *Take the logarithm:*

$$\ln\left(1+\frac{x}{N}\right)^N = N\ln\left(1+\frac{x}{N}\right).$$

With $\ln(1+u) = u - \frac{u^2}{2} + \frac{u^3}{3} - \cdots$ *and* $u = x/N$*,*

$$N\ln\left(1+\frac{x}{N}\right) = x - \frac{x^2}{2N} + \frac{x^3}{3N^2} - \cdots.$$

Every term after the first vanishes as $N\to\infty$*, so the logarithm tends to* x *and*

$$\lim_{N\to\infty}\left(1+\frac{x}{N}\right)^N = e^x.$$

Solution 2 (Other bases)**.**

$$\begin{aligned}
|+\hat{z}\rangle &\xrightarrow{S_z} \begin{pmatrix}1\\0\end{pmatrix},\\
|+\hat{z}\rangle &\xrightarrow{S_x} \begin{pmatrix}\langle+\hat{x}|+\hat{z}\rangle\\ \langle-\hat{x}|+\hat{z}\rangle\end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix},\\
|+\hat{z}\rangle &\xrightarrow{S_y} \begin{pmatrix}\langle+\hat{y}|+\hat{z}\rangle\\ \langle-\hat{y}|+\hat{z}\rangle\end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix}.
\end{aligned}$$

Solution 3 (A probability by matrix mechanics)**.**

$$\psi \xrightarrow{S_z} \frac{1}{\sqrt{5}}\begin{pmatrix}1\\2i\end{pmatrix}, \qquad \langle+\hat{x}| \xrightarrow{S_z} \frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\end{pmatrix},$$

$$\langle+\hat{x}|\psi\rangle = \frac{1+2i}{\sqrt{10}}, \qquad P\big(S_x=+\tfrac{\hbar}{2}\big) = \frac{|1+2i|^2}{10} = \frac{1}{2}.$$

Take the modulus of the whole sum, not of each term separately.

Solution 4 (The matrix of $\hat{S}_z$)**.** *Every overlap is* 0 *or* 1*, so only the diagonal survives:*

$$\langle\pm\hat{z}|\,\hat{S}_z\,|\pm\hat{z}\rangle = \pm\frac{\hbar}{2}, \qquad \hat{S}_z \xrightarrow{S_z} \frac{\hbar}{2}\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}.$$

An operator is diagonal in its own eigenbasis, with its outcomes on the diagonal.

Solution 5 (The matrix of $\hat{S}_x$)**.** *The overlaps are* $\langle +\hat{z}|\pm\hat{x}\rangle = \langle -\hat{z}|+\hat{x}\rangle = \frac{1}{\sqrt{2}}$ *and* $\langle -\hat{z}|-\hat{x}\rangle = -\frac{1}{\sqrt{2}}$*, all real. Then*

$$\langle +\hat{z}|\,\hat{S}_x\,|+\hat{z}\rangle = \frac{\hbar}{2}\left(\tfrac{1}{2}-\tfrac{1}{2}\right) = 0, \qquad \langle +\hat{z}|\,\hat{S}_x\,|-\hat{z}\rangle = \frac{\hbar}{2}\left(\tfrac{1}{2}+\tfrac{1}{2}\right) = \frac{\hbar}{2},$$
$$\langle -\hat{z}|\,\hat{S}_x\,|+\hat{z}\rangle = \frac{\hbar}{2}\left(\tfrac{1}{2}+\tfrac{1}{2}\right) = \frac{\hbar}{2}, \qquad \langle -\hat{z}|\,\hat{S}_x\,|-\hat{z}\rangle = \frac{\hbar}{2}\left(\tfrac{1}{2}-\tfrac{1}{2}\right) = 0,$$

where the + signs come from the minus in $\langle -\hat{z}|-\hat{x}\rangle$ *meeting the minus in front of the second projector. So*

$$\hat{S}_x \xrightarrow{S_z} \frac{\hbar}{2}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Solution 6 (The matrix of $\hat{S}_y$)**.** *The overlaps are complex and the bra side is conjugated:* $\langle -\hat{z}|\pm\hat{y}\rangle = \pm\frac{i}{\sqrt{2}}$, $\langle \pm\hat{y}|-\hat{z}\rangle = \mp\frac{i}{\sqrt{2}}$, *and* $\langle +\hat{z}|\pm\hat{y}\rangle = \langle \pm\hat{y}|+\hat{z}\rangle = \frac{1}{\sqrt{2}}$*. Then*

$$\langle +\hat{z}|\,\hat{S}_y\,|-\hat{z}\rangle = \frac{\hbar}{2}\left(\tfrac{1}{\sqrt{2}}\cdot\tfrac{-i}{\sqrt{2}} - \tfrac{1}{\sqrt{2}}\cdot\tfrac{i}{\sqrt{2}}\right) = -\frac{i\hbar}{2},$$
$$\langle -\hat{z}|\,\hat{S}_y\,|+\hat{z}\rangle = \frac{\hbar}{2}\left(\tfrac{i}{\sqrt{2}}\cdot\tfrac{1}{\sqrt{2}} - \tfrac{-i}{\sqrt{2}}\cdot\tfrac{1}{\sqrt{2}}\right) = +\frac{i\hbar}{2},$$

and the diagonal entries vanish because $i\cdot(-i) = 1$ *in both terms. So*

$$\hat{S}_y \xrightarrow{S_z} \frac{\hbar}{2}\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}.$$

Solution 7 (Expectation value and uncertainty by matrix mechanics)**.** *In the* S_z *basis,*

$$\psi \xrightarrow{S_z} \frac{1}{\sqrt{10}}\left[\begin{pmatrix}1\\ i\end{pmatrix} + \begin{pmatrix}2i\\ 2\end{pmatrix}\right] = \frac{1}{\sqrt{10}}\begin{pmatrix}1+2i\\ 2+i\end{pmatrix}.$$

Then

$$\langle S_z\rangle = \psi^\dagger \hat{S}_z \psi = \frac{\hbar}{2}\cdot\frac{1}{10}\begin{pmatrix}1-2i & 2-i\end{pmatrix}\begin{pmatrix}1+2i\\ -(2+i)\end{pmatrix} = \frac{\hbar}{2}\cdot\frac{5-5}{10} = 0.$$

Since $\hat{S}_z^2 = \frac{\hbar^2}{4}\mathbb{1}$, $\langle S_z^2\rangle = \frac{\hbar^2}{4}$, *so* $\Delta S_z = \hbar/2$*: both outcomes have probability* $\frac{1}{2}$ *and the spread is maximal.*

Solution 8 (Expectation value from the S_x basis)**.**

$$|\psi\rangle \xrightarrow{S_x} \psi^x = \frac{1}{\sqrt{3}}\begin{pmatrix}1\\ i\sqrt{2}\end{pmatrix}.$$

Change to the S_z *basis with* U *from Eq.* (37)*:*

$$\psi^z = U\psi^x = \frac{1}{\sqrt{6}}\begin{pmatrix}1+i\sqrt{2}\\ 1-i\sqrt{2}\end{pmatrix}, \qquad \psi^{z\dagger} = \frac{1}{\sqrt{6}}\begin{pmatrix}1-i\sqrt{2} & 1+i\sqrt{2}\end{pmatrix}.$$

Then

$$\langle S_z\rangle = \psi^{z\dagger}\hat{S}_z\psi^z = \frac{\hbar}{2}\cdot\frac{1}{6}\left(|1+i\sqrt{2}|^2 - |1-i\sqrt{2}|^2\right) = \frac{\hbar}{2}\left(\frac{3}{6}-\frac{3}{6}\right) = 0.$$

Both components have equal modulus, so both S_z outcomes have probability $\frac{1}{2}$.

Solution 9 ($\hat{S}_z$ in the S_x basis)**.** *U from Eq. (37) is real and symmetric, so $U^\dagger = U$ and*

$$\hat{S}_z \xrightarrow{S_x} U^\dagger D\, U = \frac{\hbar}{2}\cdot\frac{1}{2}\begin{pmatrix}1 & 1\\ 1 & -1\end{pmatrix}\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}\begin{pmatrix}1 & 1\\ 1 & -1\end{pmatrix} = \frac{\hbar}{2}\begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}.$$

This is $\frac{\hbar}{2}\sigma_x$: in the S_x basis, $\hat{S}_z$ looks the way $\hat{S}_x$ looks in the S_z basis.

Keeping the state in the S_x basis,

$$\langle S_z\rangle = \psi^{x\dagger}\hat{S}_z^x\psi^x = \frac{\hbar}{2}\cdot\frac{1}{3}\begin{pmatrix}1 & -i\sqrt{2}\end{pmatrix}\begin{pmatrix}i\sqrt{2}\\ 1\end{pmatrix} = \frac{\hbar}{2}\cdot\frac{i\sqrt{2}-i\sqrt{2}}{3} = 0,$$

matching the previous solution. In general, for $\psi^x = (c_+, c_-)^T$ this gives $\langle S_z\rangle = \hbar\,\mathrm{Re}\left(c_+^* c_-\right)$, which vanishes here because the relative phase is i.

Solution 10 ($R(\phi\hat{z})$ in the S_x basis)**.** *Expanding the S_x states in $\hat{z}$,*

$$R\,|\pm\hat{x}\rangle = \frac{1}{\sqrt{2}}\left(e^{-i\phi/2}\,|+\hat{z}\rangle \pm e^{i\phi/2}\,|-\hat{z}\rangle\right).$$

Hitting these with $\langle\pm\hat{x}| = \frac{1}{\sqrt{2}}\left(\langle+\hat{z}| \pm \langle-\hat{z}|\right)$ and using Euler's formula,

$$\langle+\hat{x}|\,R\,|+\hat{x}\rangle = \langle-\hat{x}|\,R\,|-\hat{x}\rangle = \frac{1}{2}\left(e^{-i\phi/2}+e^{i\phi/2}\right) = \cos\frac{\phi}{2},$$
$$\langle+\hat{x}|\,R\,|-\hat{x}\rangle = \langle-\hat{x}|\,R\,|+\hat{x}\rangle = \frac{1}{2}\left(e^{-i\phi/2}-e^{i\phi/2}\right) = -i\sin\frac{\phi}{2},$$

so

$$R(\phi\hat{z}) \xrightarrow{S_x} \begin{pmatrix}\cos\frac{\phi}{2} & -i\sin\frac{\phi}{2}\\ -i\sin\frac{\phi}{2} & \cos\frac{\phi}{2}\end{pmatrix}.$$

Check: at $\phi = \pi/2$, $R\,|+\hat{x}\rangle \xrightarrow{S_x} \frac{1}{\sqrt{2}}(1,-i)^T$, which is $|+\hat{y}\rangle$ up to a global phase, as rotating $\hat{x}$ by 90° about $\hat{z}$ should give.

B Useful Links

Stern-Gerlach experiment:
https://www.youtube.com/shorts/QG7kJXxN1Kk